

Static Holographic Boundary Theory, Precision Cosmology, and Sub-Kelvin Address-Scaled Landauer Calorimetry

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Abstract

Static Holographic Boundary Theory (SHBT) is a boundary-first closure program: the primitive object is a completed boundary conformal field theory whose torus partition function is exactly modular invariant, not a pre-existing bulk spacetime. The canonical branch $(k_\ell, k_q, K) = (26, 8, 312)$ locks the weak and color center lifts to $I_\ell^\star = 6$ and $I_q^\star = 13$, so the framing defect Δ_{fr} vanishes. On this branch, modular closure is equivalent to a vanishing bulk closure tensor and to the projected Hamiltonian constraint $\Pi_{\text{phys}} H_{\text{total}} \Pi_{\text{phys}} = 0$. The canonical closure data determine the visible modular arithmetic, the residual and completed dark ledgers $c_{\text{dark}}^{\text{res}} = 834433/362670$ and $c_{\text{dark}}^{\text{comp}} = c_{\text{dark}}^{\text{res}} + 1 = 1197103/362670$, the finite horizon register, holographic RG metric slices, baryogenesis by topological anti-baryon state decoupling, and Causal Point observer histories. The completed ledger defines the entropy-debt variable $\Delta_{\text{mod}} = c_{\text{dark}}^{\text{comp}}/24$ and an observer-frame loading law that interpolates between the Planck CMB intercept and the local Hubble value while decaying as $(1+z)^{-1}$ toward the early universe. The construction establishes the Fefferman–Graham/topological-Einstein bridge, the entropy-debt redshift ODE, the Hubble-loading amplitude A_H , the lock rate $\Gamma_{\text{lock}} = 3A_H$, the structure-growth and ISW residual equations, BBN decoupling, chronometer covariance, and the finite GET measurement rule. The finite closure analysis verifies exact branch closure, nine positive trace-normalized metric slices, nine finite-memory history entries, stress-energy-preserving topological anti-baryon state decoupling with $\eta_B = 6.449923359416 \times 10^{-10}$, Newton-lock stationarity, the unity-of-scale identity, Hubble-ladder tracking, growth suppression, the CPL mirage template, the cluster-collapse check, dark matter as the topological gravitational ghost of decoupled anti-baryons, ISW stability, and the thermodynamic light-cone entropy ledger. The theoretical construction terminates in a direct laboratory test: address-scaled Landauer calorimetry at 7 mK. For an operational capacity $C_{\text{op}} = \Delta S_{\text{payload}} + \log_2 |\mathcal{R}| + C_0$, SHBT predicts an address-dependent heat slope $k_B T \ln 2$ per address bit, whereas standard thermodynamics predicts zero slope. A superconducting ten-bit register, AuPd nanocalorimeter, explicit cryogenic noise budget, OLS/MLE decision rule, and controlled six-bit local-capacity cutoff make the observer memory postulate independently falsifiable.

1 Introduction

Quantum gravity and Standard Model phenomenology are usually formulated in opposite directions. Perturbative quantum field theory starts from local particle multiplets, gauge charges, and

renormalization-group data, and then examines the survival of this low-energy structure after coupling to gravity. Holography begins with boundary data and reconstructs local bulk spacetime and effective fields. A realistic theory must connect these directions by accounting for the observed particle content, vacuum response, baryon asymmetry, and cosmological scales without repeatedly inserting small numbers by hand. This is the naturalness problem in its broadest form. The Standard Model is phenomenologically successful, but the gravitational and vacuum sectors appear highly sensitive to ultraviolet completion unless a deeper closure principle fixes the allowed data [1, 2, 3].

A single canonical data set governs all sectors. The residual dark ledger is $c_{\text{dark}}^{\text{res}} = 834433/362670$, and the completed finite-screen ledger entering the Noether bridge and precision cosmology is $c_{\text{dark}}^{\text{comp}} = c_{\text{dark}}^{\text{res}} + 1 = 1197103/362670$. The retained D_5 boundary-cell measure is denoted by κ_{D_5} . In the precision-cosmology sector, $\Delta_{\text{mod}} = c_{\text{dark}}^{\text{comp}}/24$, A_H is the Hubble-loading amplitude, and $\Gamma_{\text{lock}} = 3A_H$ is the Causal Point lock rate. The microscopic center-lift, completed-ledger, and boundary-measure definitions are collected in Appendices A–B.

Static Holographic Boundary Theory (SHBT) proposes that the missing closure principle is exact modular invariance of a completed boundary conformal field theory (CFT). The starting point is neither a fluctuating classical spacetime nor a pre-existing cosmological history, but a completed boundary Hilbert space and a physical torus partition function

$$Z_{\partial}(\tau, \bar{\tau}) = \sum_{I,J} M_{IJ} \chi_I(\tau) \overline{\chi_J(\tau)}, \quad [M, S_{\partial}] = [M, T_{\partial}] = 0, \quad M_{IJ} \in \mathbb{Z}_{\geq 0}. \quad (1)$$

The role of modular covariance follows the standard lesson that torus partition functions in two-dimensional CFT are not arbitrary sums of characters: their operator content is constrained by the modular group [4, 5]. SHBT strengthens this consistency condition into a first principle. A boundary sector is physical only if it closes under the completed modular kernels, and any residual mismatch is encoded in a scalar framing defect Δ_{fr} . The central claim is the closure chain

$$[M, S_{\partial}] = [M, T_{\partial}] = 0 \iff \Delta_{\text{fr}} = 0 \iff \mathcal{E}_{\mu\nu} = 0 \iff \Pi_{\text{phys}} H_{\text{total}} \Pi_{\text{phys}} = 0. \quad (2)$$

Thus holographic closure is not imposed after a bulk theory has been guessed. Rather, the vanishing of the bulk closure tensor and the disappearance of an independent external time generator are the bulk and Hamiltonian images of completed modular closure on the boundary. On the finite visible block this statement is tested directly by

$$\|[M, S_{\partial}]\|_F = 0, \quad \|[M, T_{\partial}]\|_F = 0, \quad \Delta_{\text{fr}} = 0, \quad (3)$$

together with equality of the torus partition function under the two modular generators. The implementation correspondence for these physical conditions is collected separately in Section 12.

The canonical branch studied here is

$$\mathfrak{b}_* = (k_{\ell}, k_q, K) = (26, 8, 312), \quad I_{\ell}^* = \frac{K}{2k_{\ell}} = 6, \quad I_q^* = \frac{K}{3k_q} = 13. \quad (4)$$

Within SHBT this branch is the anomaly-free fixed point because the lepton, color, and parent completion levels lock to integers and give

$$\Delta_{\text{fr}}(312, 26, 8) = \max\left(\left\|\frac{312}{2 \cdot 26}\right\|_{\mathbb{Z}}, \left\|\frac{312}{3 \cdot 8}\right\|_{\mathbb{Z}}\right) = 0. \quad (5)$$

The visible factors are $SU(2)_{26}$ and $SU(3)_8$, embedded in an $SO(10)_{312}$ parent completion and a dark completion sector. The dark sector is therefore not an optional phenomenological reservoir. It is the finite ledger required by exact closure. The residual part is

$$c_{\text{dark}}^{\text{res}} = \frac{834433}{362670} \simeq 2.300805139658643, \quad 8\pi G_N = \frac{12}{c_{\text{dark}}^{\text{res}} M_P^2}. \quad (6)$$

The finite-screen cosmology construction uses the completed ledger

$$c_{\text{dark}}^{\text{comp}} = c_{\text{dark}}^{\text{res}} + 1 = \frac{1197103}{362670} \simeq 3.300805139658643, \quad \Delta_{\text{mod}} = \frac{c_{\text{dark}}^{\text{comp}}}{24}. \quad (7)$$

These quantities jointly specify the visible affine levels, their parent embedding indices, the residual and completed dark ledgers, and the vanishing framing obstruction. No additional continuous parameter enters this branch arithmetic.

The construction has four linked components. First, the completed boundary CFT is treated as the primary object,

$$\mathcal{H}_{\partial}^{\text{comp}} = \mathcal{H}_{SU(2)_{26}} \otimes \mathcal{H}_{SU(3)_8} \otimes \mathcal{H}_{SO(10)_{312}} \otimes \mathcal{H}_{\text{dark}}, \quad (8)$$

whose physical spectrum is the modular-invariant, nonnegative-integer pairing of completed characters. This is the algebraic source of gauge and gravitational consistency. Second, entropy self-resolution turns the finite boundary register

$$N = N_{\text{sat}} = \frac{3\pi}{L_P^2 \Lambda_{\text{holo}}}, \quad \Lambda_{\text{holo}} > 0, \quad (9)$$

into normalized loading and entanglement densities. Their entropy gradients define a holographic renormalization-group flow into bulk metric slices. This extends the holographic and black-hole thermodynamic framework [7, 8] by making the entropy budget finite and directly testable through the symmetry, trace normalization, and positive definiteness of every projected metric slice.

Third, SHBT reframes baryogenesis as topological anti-baryon state decoupling. The observed baryon asymmetry is usually framed as a dynamical departure from baryon number, C, CP, and equilibrium conditions [9]. Here the active render charge is separated from passive stress-energy. Anti-baryon visible charges are projected out of the actively rendered gauge channels, while the passive gravitational source is preserved in the dark completion sector. The resulting asymmetry is therefore an observer-visible render imbalance, not a failure of completed stress-energy accounting. The corresponding physical checks are

- the exact baryogenesis identity for $C_{\text{sph}} J_{\text{CP}}^{\text{topo}} M_N / M_P$,
- the charge-decoupling map on active anti-baryon gauge channels,
- equality of the passive stress-energy source before and after decoupling, and
- the comparison of visible render content at fixed gravitational accounting.

The resulting canonical value is $\eta_B = 6.449923359416 \times 10^{-10}$.

The same state-decoupling mechanism predicts dark matter as the passive gravitational ghost of the stripped anti-baryons. The canonical-branch calculation yields $\Omega_{\text{DM}}/\Omega_b \simeq 5.34$, consistent with the observed ratio $\simeq 5.4$.

Fourth, the Causal Point makes the observer part of the same finite register. SHBT does not assume an external classical observer with unlimited memory. An observer is a finite local address equipped with a light-cone sample, a local property packet, a hidden-state budget, and a history projection operator $\Pi_{A,\ell}$. A measurement is admitted only when the residual entropy budget is nonnegative. Observable time and crystallized history are therefore finite-register projections rather than primitive background parameters. The required physical data are the past-light-cone sample, the local property packet, the coordinate-log entry, and the entropy-budget inequality defining the admissible history projection.

The main results follow from the same closure data. Because the branch $(26, 8, 312)$ has exact framing closure, the bulk closure tensor vanishes and the topological Einstein equation follows in the SHBT sector:

$$\mathcal{E}_{\mu\nu} = \frac{c_{\text{dark}}^{\text{res}} M_P^2}{12} (G_{\mu\nu} + \Lambda_{\text{holo}} g_{\mu\nu}) - T_{\mu\nu}^{\text{vis}} = 0. \quad (10)$$

The static physical sector has no independent external time generator because the completed Hamiltonian constraint annihilates physical states, $\Pi_{\text{phys}} H_{\text{total}} \Pi_{\text{phys}} = 0$. Finite-register entropy flow produces symmetric, trace-normalized, positive metric slices and an observer-frame effective cosmology. The Causal Point construction turns measurement into an entropy-budgeted projection with explicit failure conditions. Topological anti-baryon state decoupling yields the benchmark baryon asymmetry while preserving passive stress-energy. The complete closure ledger therefore consists of the canonical branch, $\Delta_{\text{fr}} = 0$, exact modular commutation, the projected zero-energy constraint, the finite bit budget, nine bulk metric slices, nine crystallized history entries, $\eta_B = 6.449923359416 \times 10^{-10}$, and stress-energy preservation. Reproduction commands and machine-readable accessors are deferred to Section 12.

The calculations constitute a self-consistency test of the boundary-first closure program. The reported numerical values—including the baryon asymmetry $\eta_B = 6.449923359416 \times 10^{-10}$, the dark completion residue $c_{\text{dark}}^{\text{comp}} = 1197103/362670$, and the finite horizon register $N_{\text{sat}} \approx 3.31 \times 10^{122}$ —are exact canonical-branch retrodictions rather than independent empirical inputs. They follow from modular invariance of the completed boundary CFT and the associated finite-register closure conditions. The finite-dimensional checks establish that the canonical branch is algebraically closed and that its bulk and observer projections yield the reported scales. In particular, the measured baryon asymmetry is compared with a symmetry-derived value rather than used to adjust a free parameter. Any displacement from the locked branch reopens the framing defect Δ_{fr} and breaks the closure chain. Independent tests must therefore target signatures not used in the branch construction, including time dependence in $w(a)$, CMB non-Gaussianity, and direct dark-matter channels.

The relation between SHBT and existing approaches is precise. Like AdS/CFT, SHBT treats a boundary quantum theory as sufficient data for a bulk gravitational description [10, 11, 12]. Unlike standard AdS/CFT, the present construction is not organized around a large- N conformal gauge theory with an asymptotically anti-de Sitter bulk. It is organized around a completed static boundary CFT, modular closure, and a finite holographic register. Like entropic-gravity programs, SHBT treats geometry as downstream of information and entropy [16]; unlike a pure entropic-force derivation, its entropy gradients are constrained by modular closure and checked slice-by-slice by

the metric closure conditions. Like topological and constraint-based approaches to quantum gravity, SHBT removes an external time generator from the physical sector; unlike a purely formal constraint algebra, it ties that removal to visible-sector branch arithmetic, dark completion, baryogenesis, and observer memory.

The paper is organized as follows. Section 2 derives the canonical branch and states the axiomatic closure conditions. Section 3 constructs the modular data and boundary partition function. Section 4 defines the finite entropy densities that self-resolve the boundary register, and Section 5 maps their gradients into bulk metric slices. Section 6 develops topological anti-baryon state decoupling and stress-energy preservation, while Section 7 defines Causal Points and history crystallization. Section 8 collects the finite-dimensional verification results, Section 9 derives the precision cosmology consequences, and Section 10 converts the Causal Point GET rule into a sub-10 mK calorimetric falsification experiment. Section 11 synthesizes the closure structure and its remaining limitations. Throughout, the analytic derivation and the numerical closure diagnostics are kept in parallel so that the same branch, defect, entropy budget, metric projection, Causal Point, and baryogenesis identity must agree. Their exact software mapping is isolated in Section 12.

2 Canonical Branch and Axiomatic Foundation

The canonical branch is derived by treating the completed boundary theory, not a bulk Lagrangian, as the primitive object. For arbitrary visible levels (k_ℓ, k_q) and parent level K , the completed boundary Hilbert space is postulated to factor as

$$\mathcal{H}_\partial^{\text{comp}}(k_\ell, k_q, K) = \mathcal{H}_{SU(2)_{k_\ell}} \otimes \mathcal{H}_{SU(3)_{k_q}} \otimes \mathcal{H}_{SO(10)_K} \otimes \mathcal{H}_{\text{dark}}. \quad (11)$$

The collective primary index is written $I = (i_\ell, i_q, i_{10}, \alpha)$, and its completed character is

$$\chi_I(\tau) = \chi_{i_\ell}^{(k_\ell)}(\tau) \chi_{i_q}^{(k_q)}(\tau) \chi_{i_{10}}^{(K)}(\tau) \chi_\alpha^{\text{dark}}(\tau). \quad (12)$$

The physical torus partition function has the form

$$Z_\partial(\tau, \bar{\tau}) = \sum_{I, J} M_{IJ} \chi_I(\tau) \overline{\chi_J(\tau)}, \quad M_{IJ} \in \mathbb{Z}_{\geq 0}, \quad (13)$$

with nonnegative integer multiplicities. The first axiom is that the completed pairing matrix is invariant under the full boundary modular kernels,

$$[M, S_\partial] = 0, \quad [M, T_\partial] = 0, \quad (14)$$

where

$$\begin{aligned} S_\partial &= S_{SU(2)_{k_\ell}} \otimes S_{SU(3)_{k_q}} \otimes S_{SO(10)_K} \otimes S_{\text{dark}}, \\ T_\partial &= T_{SU(2)_{k_\ell}} \otimes T_{SU(3)_{k_q}} \otimes T_{SO(10)_K} \otimes T_{\text{dark}}. \end{aligned} \quad (15)$$

Equivalently, $Z_\partial(\tau + 1, \bar{\tau} + 1) = Z_\partial(\tau, \bar{\tau})$ and $Z_\partial(-1/\tau, -1/\bar{\tau}) = Z_\partial(\tau, \bar{\tau})$, so the allowed operator content is a modular-invariant completed pairing rather than an arbitrary list of boundary primaries [4, 5].

The integral-spin condition is the local form of the T -constraint. A character transforms as

$$\chi_I(\tau + 1) = \exp\left[2\pi i \left(h_I - \frac{c_I}{24}\right)\right] \chi_I(\tau), \quad (16)$$

while the anti-holomorphic character contributes the conjugate phase. Therefore the summand with coefficient M_{IJ} is T -invariant only if

$$M_{IJ} \neq 0 \implies \exp\left\{2\pi i \left[h_I - h_J - \frac{c_I - c_J}{24}\right]\right\} = 1, \quad (17)$$

or, equivalently,

$$\boxed{M_{IJ} \neq 0 \implies h_I - h_J - \frac{c_I - c_J}{24} \in \mathbb{Z}.} \quad (18)$$

This condition is the boundary algebraic precursor of the SHBT framing defect. The visible conformal weights entering the finite block are

$$\begin{aligned} h_a^{SU(2)_k} &= \frac{a(a+2)}{4(k+2)}, \\ h_{(p,q)}^{SU(3)_k} &= \frac{p^2 + q^2 + pq + 3p + 3q}{3(k+3)}, \end{aligned} \quad (19)$$

with central charges

$$c_{SU(2)_k} = \frac{3k}{k+2}, \quad c_{SU(3)_k} = \frac{8k}{k+3}, \quad c_{SO(10)_K} = \frac{45K}{K+8}. \quad (20)$$

The canonical branch is the minimal parent completion compatible with the visible lepton and quark levels used throughout the analysis. The embedding locks require the parent level to close simultaneously over the lepton and color descendants,

$$I_\ell = \frac{K}{2k_\ell} \in \mathbb{Z}, \quad I_q = \frac{K}{3k_q} \in \mathbb{Z}. \quad (21)$$

For the visible levels $k_\ell = 26$ and $k_q = 8$, the smallest positive parent level satisfying both locks is

$$K = \text{lcm}(2 \cdot 26, 3 \cdot 8) = \text{lcm}(52, 24) = 312. \quad (22)$$

Thus

$$\mathfrak{b}_* = (k_\ell, k_q, K) = (26, 8, 312), \quad I_\ell^* = \frac{312}{2 \cdot 26} = 6, \quad I_q^* = \frac{312}{3 \cdot 8} = 13. \quad (23)$$

This simultaneously fixes the two visible affine levels, the minimal parent completion level, and the two integer embedding indices. The least-common-multiple construction makes the result unique among positive parent levels.

The scalar obstruction to these integer locks is the framing defect

$$\boxed{\Delta_{\text{fr}}(K, k_\ell, k_q) = \max\left(\left\|\frac{K}{2k_\ell}\right\|_{\mathbb{Z}}, \left\|\frac{K}{3k_q}\right\|_{\mathbb{Z}}\right),} \quad \|x\|_{\mathbb{Z}} = \min_{n \in \mathbb{Z}} |x - n|. \quad (24)$$

On the canonical branch both arguments are integers, so

$$\Delta_{\text{fr}}(312, 26, 8) = \max(\|6\|_{\mathbb{Z}}, \|13\|_{\mathbb{Z}}) = 0. \quad (25)$$

The branch is therefore anomaly-free in the SHBT sense: the parent completion has no residual visible framing mismatch, and the scalar closure obstruction vanishes exactly rather than only within numerical tolerance.

The second axiom identifies the bulk closure tensor as the covariant image of the scalar modular obstruction. The standard holographic bridge begins with the Fefferman–Graham expansion of an asymptotically locally AdS metric [6, 15],

$$ds^2 = \frac{L^2}{z^2} \left(dz^2 + g_{ij}(x, z) dx^i dx^j \right), \quad z \rightarrow 0, \quad (26)$$

where

$$g_{ij}(x, z) = g_{(0)ij}(x) + z^2 g_{(2)ij}(x) + \cdots + z^d g_{(d)ij}(x) + z^d \log z^2 h_{(d)ij}(x) + \cdots. \quad (27)$$

The boundary metric $g_{(0)ij}$ is the source for the CFT stress tensor, and holographic renormalization reads the renormalized one-point function from the normalizable coefficient,

$$\langle T_{ij} \rangle = \frac{dL^{d-1}}{16\pi G_N} g_{(d)ij} + T_{ij}^{\text{ct}}[g_{(0)}], \quad (28)$$

with local counterterms fixed by the same boundary data. Under a boundary Weyl variation,

$$\delta_\sigma W_{\text{CFT}}[g_{(0)}] = \int d^d x \sqrt{g_{(0)}} \sigma \mathcal{A}_{\text{Weyl}}, \quad \langle T^i_i \rangle = \mathcal{A}_{\text{Weyl}}. \quad (29)$$

In SHBT the framing part of this anomaly is not an independent local curvature functional. It is the scalar obstruction left by the completed modular pairing:

$$\mathcal{A}_{\text{Weyl}}^{\text{fr}} = d \mathcal{Q}_{\text{fr}} \Delta_{\text{fr}}, \quad \mathcal{E}_{ij}^\partial = \mathcal{Q}_{\text{fr}} \Delta_{\text{fr}} g_{(0)ij}. \quad (30)$$

The unique covariant bulk continuation of this purely scalar boundary obstruction on a closed branch is therefore metric-proportional,

$$\boxed{\mathcal{E}_{\mu\nu} = \mathcal{Q}_{\text{fr}} \Delta_{\text{fr}} g_{\mu\nu}}, \quad \mathcal{Q}_{\text{fr}} \neq 0. \quad (31)$$

Equivalently, the framing defect is a topological Weyl-anomaly source: if the integer center lifts close, the source vanishes; if they do not, the only allowed covariant obstruction is proportional to the metric.

The same tensor is defined by the completed SHBT stress accounting,

$$\mathcal{E}_{\mu\nu} \equiv \frac{c_{\text{dark}}^{\text{res}} M_P^2}{12} (G_{\mu\nu} + \Lambda_{\text{holo}} g_{\mu\nu}) - T_{\mu\nu}^{\text{vis}}, \quad 8\pi G_N = \frac{12}{c_{\text{dark}}^{\text{res}} M_P^2}. \quad (32)$$

Combining Eqs. (31) and (32) gives the off-branch topological Einstein equation with a framing source,

$$G_{\mu\nu} + \Lambda_{\text{holo}} g_{\mu\nu} = \frac{12}{c_{\text{dark}}^{\text{res}} M_P^2} \left(T_{\mu\nu}^{\text{vis}} + \mathcal{Q}_{\text{fr}} \Delta_{\text{fr}} g_{\mu\nu} \right). \quad (33)$$

On the canonical branch, Eq. (25) gives $\Delta_{\text{fr}} = 0$, hence

$$\Delta_{\text{fr}} = 0 \iff \mathcal{E}_{\mu\nu} = 0. \quad (34)$$

Using Eq. (32), the vanishing condition is precisely

$$G_{\mu\nu} + \Lambda_{\text{holo}} g_{\mu\nu} = \frac{12}{c_{\text{dark}}^{\text{res}} M_P^2} T_{\mu\nu}^{\text{vis}} = 8\pi G_N T_{\mu\nu}^{\text{vis}}. \quad (35)$$

Thus the complete boundary-to-bulk derivation chain is

$$\begin{aligned} [M, S_{\partial}] = [M, T_{\partial}] = 0 &\implies h_I - h_J - \frac{c_I - c_J}{24} \in \mathbb{Z} \implies \frac{K}{2k_{\ell}}, \frac{K}{3k_q} \in \mathbb{Z} \implies \Delta_{\text{fr}} = 0, \\ \Delta_{\text{fr}} = 0 &\implies \mathcal{A}_{\text{Weyl}}^{\text{fr}} = 0 \implies \mathcal{E}_{\mu\nu} = 0 \implies G_{\mu\nu} + \Lambda_{\text{holo}} g_{\mu\nu} = 8\pi G_N T_{\mu\nu}^{\text{vis}}. \end{aligned} \quad (36)$$

The tensor equation is therefore not an independent tuning condition; it is the bulk representative of exact modular closure.

The final axiom is static global Hamiltonian cancellation. Let

$$H_{\text{total}} = H_{\partial} + H_{\text{bulk}} + H_{\text{stabilizer}}, \quad (37)$$

where H_{∂} acts on the completed boundary characters, H_{bulk} is the bulk image of the closed boundary data, and $H_{\text{stabilizer}}$ enforces the closure syndromes. The physical projector is defined by the simultaneous constraints

$$\Pi_{\text{phys}} M = M \Pi_{\text{phys}} = M, \quad P_{\text{fr}} \Pi_{\text{phys}} = 0, \quad \mathcal{E}_{\mu\nu} \Pi_{\text{phys}} = 0. \quad (38)$$

On this sector the stabilizer generator is fixed, not freely chosen:

$$H_{\text{stabilizer}} \Pi_{\text{phys}} = -(H_{\partial} + H_{\text{bulk}}) \Pi_{\text{phys}}. \quad (39)$$

Multiplying Eq. (37) by Π_{phys} on both sides gives

$$\Pi_{\text{phys}} H_{\text{total}} \Pi_{\text{phys}} = 0. \quad (40)$$

Therefore, for every physical completed-boundary state satisfying $|\Psi_{\text{phys}}\rangle = \Pi_{\text{phys}} |\Psi_{\text{phys}}\rangle$,

$$\boxed{H_{\text{total}} |\Psi_{\text{phys}}\rangle = 0.} \quad (41)$$

This is the static-block statement that the completed physical sector has no independent external time generator.

The axiomatic content of this section is therefore the simultaneous closure of the visible modular blocks, the scalar framing obstruction, the topological Einstein relation, and the projected Hamiltonian constraint. The exact software realization of each member of this chain is documented only in Section 12.

3 Modular Data and Boundary Partition Function

This section makes explicit the modular data used by the completed boundary partition function. The general Wess–Zumino–Witten rule is that an affine factor G_k has integrable highest weights λ , modular character vector χ^{G_k} , dual Coxeter number $g^\vee$, central charge

$$c_{G_k} = \frac{k \dim G}{k + g^\vee}, \quad (42)$$

and diagonal modular- T entries

$$T_{\lambda\mu}^{G_k} = \delta_{\lambda\mu} \exp \left[2\pi i \left(h_\lambda^{G_k} - \frac{c_{G_k}}{24} \right) \right], \quad h_\lambda^{G_k} = \frac{(\lambda, \lambda + 2\rho)}{2(k + g^\vee)}. \quad (43)$$

Here ρ is the Weyl vector and the inner product is normalized so long roots have length squared 2. For commutator tests it is sufficient to use the reduced visible phase $\exp(2\pi i h_\lambda)$; the omitted scalar factor $\exp(-2\pi i c/24)$ is common to a fixed factor and cancels between the holomorphic and anti-holomorphic sectors, and it also leaves the commutator norms unchanged.

For $SU(2)_k$, the integrable labels are $a = 0, 1, \dots, k$. With spin $j = a/2$ and $C_2(j) = j(j+1) = a(a+2)/4$, Eq. (43) gives

$$h_a^{SU(2)_k} = \frac{a(a+2)}{4(k+2)}, \quad c_{SU(2)_k} = \frac{3k}{k+2}. \quad (44)$$

The A_1 Weyl sum has two terms and reduces to the sine kernel

$$S_{ab}^{SU(2)_k} = \sqrt{\frac{2}{k+2}} \sin \left(\frac{\pi(a+1)(b+1)}{k+2} \right), \quad T_{ab}^{SU(2)_k} = \delta_{ab} \exp \left[2\pi i \left(h_a^{SU(2)_k} - \frac{c_{SU(2)_k}}{24} \right) \right]. \quad (45)$$

For $SU(3)_k$, the integrable labels are Dynkin pairs (p, q) with $p, q \geq 0$ and $p+q \leq k$. The quadratic Casimir is

$$C_2(p, q) = \frac{p^2 + q^2 + pq + 3p + 3q}{3}, \quad (46)$$

so that

$$h_{(p,q)}^{SU(3)_k} = \frac{p^2 + q^2 + pq + 3p + 3q}{3(k+3)}, \quad c_{SU(3)_k} = \frac{8k}{k+3}. \quad (47)$$

It is useful to embed a Dynkin label in the traceless three-dimensional weight space by

$$v(p, q) = \left(\frac{2p+q}{3}, \frac{-p+q}{3}, -\frac{p+2q}{3} \right), \quad \rho = v(1, 1) = (1, 0, -1). \quad (48)$$

Then the relevant A_2 Weyl-sum formula is

$$S_{(p,q),(r,s)}^{SU(3)_k} = \frac{i^3}{\sqrt{3}(k+3)} \sum_{\sigma \in S_3} \text{sgn}(\sigma) \exp \left[-\frac{2\pi i}{k+3} (\sigma(v(p, q) + \rho), v(r, s) + \rho) \right], \quad (49)$$

with

$$T_{(p,q),(r,s)}^{SU(3)_k} = \delta_{(p,q),(r,s)} \exp \left[2\pi i \left(h_{(p,q)}^{SU(3)_k} - \frac{c_{SU(3)_k}}{24} \right) \right]. \quad (50)$$

For the parent completion $SO(10)_K$, the Lie algebra is D_5 , with $g^\vee = 8$, rank 5, $\dim SO(10) = 45$, and 20 positive roots. Thus

$$c_{SO(10)_K} = \frac{45K}{K+8}. \quad (51)$$

If Λ and M are integrable D_5 highest weights, the corresponding formal kernels are

$$\begin{aligned} S_{\Lambda M}^{SO(10)_K} &= \frac{i^{20}}{2(K+8)^{5/2}} \sum_{w \in W(D_5)} \text{sgn}(w) \exp \left[-\frac{2\pi i}{K+8} (w(\Lambda + \rho), M + \rho) \right], \\ T_{\Lambda M}^{SO(10)_K} &= \delta_{\Lambda M} \exp \left[2\pi i \left(\frac{(\Lambda, \Lambda + 2\rho)}{2(K+8)} - \frac{45K}{24(K+8)} \right) \right]. \end{aligned} \quad (52)$$

The finite visible-block analysis does not require enumeration of the full $SO(10)_{312}$ character table; instead, this parent factor enters through the branch locks, the framing-defect condition, and the central charge in Eq. (51).

The affine data on the completed canonical branch are summarized in Table 1. The visible central charge is the sum of the first two visible factors; the completed ledger is the finite-screen dark contribution used by the Noether bridge and cosmology. The first two entries follow from the affine

Affine factor	Level	Central charge	Center data
$SU(2)_{k_\ell}$	$k_\ell = 26$	$\frac{39}{14} = 2.785714285714$	$\mathbb{Z}_2$ spin parity
$SU(3)_{k_q}$	$k_q = 8$	$\frac{64}{11} = 5.818181818182$	$\mathbb{Z}_3$ color triality
$SO(10)_K$	$K = 312$	$\frac{351}{8} = 43.875000000000$	$\mathbb{Z}_4$ spinor center
$\mathcal{H}_{\text{dark}}$	—	$c_{\text{dark}}^{\text{comp}} = \frac{1197103}{362670} = 3.300805139659$	—

Table 1: Affine algebraic parameters and central charges of the completed boundary CFT factors on the canonical branch.

Wess–Zumino–Witten central-charge formula, the third from the D_5 parent algebra, and the last from the exact completed-ledger arithmetic derived in Appendix B.4.

On the canonical branch, the resolved visible block is the 3×3 slice

$$\begin{aligned} \mathcal{A}_{\text{vis}} &= (k_\ell - 4, k_\ell - 3, k_\ell) = (22, 23, 26), \\ \mathcal{W}_{\text{vis}} &= ((0, 0), (1, 0), (0, 1)). \end{aligned} \quad (53)$$

This slice is not the full completed Hilbert space. It is the visible finite-dimensional block on which the modular kernels and torus pairing are evaluated. The combined visible index is

$$A = (i, j), \quad a_i \in (22, 23, 26), \quad w_j \in ((0, 0), (1, 0), (0, 1)), \quad (54)$$

so the visible space has 9 basis entries. In that ordering,

$$S_{\partial}^{\text{vis}} = \text{Re } S_{SU(2)_{26}}^{\text{vis}} \otimes \text{Re } S_{SU(3)_8}^{\text{vis}}, \quad T_{\partial}^{\text{vis}} = \text{diag} \left(T_{SU(2)_{26}}^{\text{vis}} \otimes T_{SU(3)_8}^{\text{vis}} \right). \quad (55)$$

The benchmark central charges and conformal weights are

$$\begin{aligned}
c_{SU(2)_{26}} &= \frac{39}{14} = 2.785714285714, & c_{SU(3)_8} &= \frac{64}{11} = 5.818181818182, \\
c_{SO(10)_{312}} &= \frac{351}{8} = 43.875, & c_{\text{vis}} &= \frac{1325}{154} = 8.603896103896, \\
h_{22}^{SU(2)_{26}} &= \frac{33}{7} = 4.714285714286, & h_{23}^{SU(2)_{26}} &= \frac{575}{112} = 5.133928571429, \\
h_{26}^{SU(2)_{26}} &= \frac{13}{2} = 6.5, & h_{(0,0)}^{SU(3)_8} &= 0, \\
h_{(1,0)}^{SU(3)_8} &= \frac{4}{33} = 0.121212121212, & h_{(0,1)}^{SU(3)_8} &= \frac{4}{33} = 0.121212121212.
\end{aligned} \tag{56}$$

The resulting visible S blocks are

$$\begin{aligned}
S_{SU(2)_{26}}^{\text{vis}} &\simeq \begin{pmatrix} 0.088270792276 & -0.208953252971 & 0.142191553507 \\ -0.208953252971 & 0.260560444585 & -0.115960306962 \\ 0.142191553507 & -0.115960306962 & 0.029923764933 \end{pmatrix}, \\
\text{Re } S_{SU(3)_8}^{\text{vis}} &\simeq \begin{pmatrix} 0.018018539201 & 0.048334858719 & 0.048334858719 \\ 0.048334858719 & 0.110479936109 & 0.110479936109 \\ 0.048334858719 & 0.110479936109 & 0.110479936109 \end{pmatrix}.
\end{aligned} \tag{58}$$

The imaginary parts of the displayed $SU(3)$ entries are at roundoff scale for the first row and are conjugate in the fundamental/anti-fundamental subblock; the real restriction shown above defines the finite visible block. The reduced visible T phases are

$$\begin{aligned}
T_{SU(2)_{26}}^{\text{vis}} &\simeq \begin{pmatrix} -0.222520933956 - 0.974927912182i \\ 0.666346577952 + 0.745642164883i \\ -1.000000000000 \end{pmatrix}, \\
T_{SU(3)_8}^{\text{vis}} &\simeq \begin{pmatrix} 1 \\ 0.723734038105 + 0.690079011482i \\ 0.723734038105 + 0.690079011482i \end{pmatrix}.
\end{aligned} \tag{59}$$

The completed boundary partition function is the torus pairing of completed characters,

$$Z_{\partial}(\tau, \bar{\tau}) = \sum_{I,J} M_{IJ} \chi_I(\tau) \overline{\chi_J(\tau)}, \quad M_{IJ} \in \mathbb{Z}_{\geq 0}, \tag{60}$$

where $I = (a, (p, q), \Lambda, \alpha)$ includes the $SU(2)$ label, $SU(3)$ label, $SO(10)$ parent label, and dark-sector label. In vector notation this is $Z_{\partial} = \chi^{\dagger} M \chi$, up to the conventional placement of the holomorphic index. Since the character vector transforms as

$$\chi(\tau + 1) = T_{\partial} \chi(\tau), \quad \chi(-1/\tau) = S_{\partial} \chi(\tau), \tag{61}$$

modular invariance follows from

$$[M, T_{\partial}] = 0, \quad [M, S_{\partial}] = 0. \tag{62}$$

Indeed,

$$Z_{\partial}(\tau + 1, \bar{\tau} + 1) = \chi(\tau)^{\dagger} T_{\partial}^{\dagger} M T_{\partial} \chi(\tau) = \chi(\tau)^{\dagger} M \chi(\tau), \tag{63}$$

$$Z_{\partial}(-1/\tau, -1/\bar{\tau}) = \chi(\tau)^{\dagger} S_{\partial}^{\dagger} M S_{\partial} \chi(\tau) = \chi(\tau)^{\dagger} M \chi(\tau), \tag{64}$$

using unitarity of the modular kernels. Therefore

$$\boxed{Z_{\partial}(\tau + 1, \bar{\tau} + 1) = Z_{\partial}(\tau, \bar{\tau}), \quad Z_{\partial}(-1/\tau, -1/\bar{\tau}) = Z_{\partial}(\tau, \bar{\tau}).} \quad (65)$$

The completed pairing matrices are therefore classified by the finite commutant of the two modular generators,

$$\mathfrak{M}_{\partial} = \{M \in \text{Mat}_N(\mathbb{Z}_{\geq 0}) \mid M_{00} = 1, [M, S_{\partial}] = 0, [M, T_{\partial}] = 0\}. \quad (66)$$

Writing

$$\theta_I = \exp\left[2\pi i \left(h_I - \frac{c_I}{24}\right)\right], \quad (T_{\partial})_{IJ} = \theta_I \delta_{IJ}, \quad (67)$$

the T commutator is

$$([M, T_{\partial}])_{IJ} = M_{IJ}(\theta_J - \theta_I). \quad (68)$$

Hence every nonzero pairing lies inside a common modular-spin sector,

$$M_{IJ} \neq 0 \implies \theta_I = \theta_J \iff h_I - h_J - \frac{c_I - c_J}{24} \in \mathbb{Z}. \quad (69)$$

This is the integral-spin closure condition of the completed CFT. The canonical locks in Eq. (23) refine this condition by resolving the visible center lifts: an admissible visible pairing must preserve both the $SU(2)$ spin-parity lift and the $SU(3)$ color-triality lift into the $SO(10)_{312}$ parent. Pairings that mix different lifted center classes would change one of the integer quotients $K/(2k_{\ell})$ or $K/(3k_q)$ and therefore contribute directly to Δ_{fr} .

On the 9-state visible block this leaves only pointwise pairings after the center-lift classes have been resolved. Thus the visible restriction has the form

$$M_{\text{vis}} = \text{diag}(m_1, \dots, m_9), \quad m_A \in \mathbb{Z}_{\geq 0}. \quad (70)$$

The remaining S constraint forces all diagonal multiplicities to coincide:

$$([M_{\text{vis}}, S_{\partial}^{\text{vis}}])_{AB} = (m_A - m_B)(S_{\partial}^{\text{vis}})_{AB} = 0. \quad (71)$$

Every entry of the displayed benchmark $S_{\partial}^{\text{vis}}$ is nonzero because the visible $SU(2)_{26}$ and real $SU(3)_8$ subblocks in Eq. (58) have no zero entries. Therefore $m_A = m_B$ for all A, B , and the vacuum normalization $M_{00} = 1$ fixes the common value to one:

$$\boxed{M_{\text{vis}} = \mathbb{1}_9.} \quad (72)$$

This is the unique center-resolved visible-block representative in $\mathfrak{M}_{\partial}$. It keeps the visible sector pointwise paired at the self-dual point $\tau = i$ and avoids adding a spin-parity or color-triality framing defect.

The finite visible-block partition test takes $M_{\text{vis}} = \mathbb{1}_9$ and the leading visible character monomials

$$\chi_{(i,j)}^{\text{vis}}(\tau) = q^{h_{a_i}^{SU(2)_{26}} + h_{w_j}^{SU(3)_8} - c_{\text{vis}}/24}, \quad q = e^{2\pi i \tau}, \quad c_{\text{vis}} = \frac{1325}{154}. \quad (73)$$

Their diagonal torus pairing is

$$Z_{\partial}^{\text{vis}}(\tau) = \overline{\chi^{\text{vis}}(\tau)}^T M_{\text{vis}} \chi^{\text{vis}}(\tau). \quad (74)$$

For $\tau = i$ the numerical evaluation gives

$$\begin{aligned} Z_{\partial}^{\text{vis}}(i) &= 2.441381789163 \times 10^{-24}, \\ Z_{\partial}^{\text{vis}}(i+1) &= 2.441381789163 \times 10^{-24}, \end{aligned} \tag{75}$$

$$Z_{\partial}^{\text{vis}}(-1/i) = 2.441381789163 \times 10^{-24}. \tag{76}$$

The equality under the S move in this numerical line is pointwise because i is the self-dual point of $\tau \mapsto -1/\tau$; the nontrivial finite-block test of the two modular generators is the matrix commutator condition. The complete visible-block closure ledger is

Physical closure observable	Theoretical value
$\ [M_{\text{vis}}, S_{\partial}^{\text{vis}}]\ _F$	0
$\ [M_{\text{vis}}, T_{\partial}^{\text{vis}}]\ _F$	0
Modular invariance of $Z_{\partial}^{\text{vis}}$	satisfied
Integral-spin closure	satisfied
Combined boundary closure	satisfied

Thus the visible boundary block satisfies

$$\|M_{\text{vis}} S_{\partial}^{\text{vis}} - S_{\partial}^{\text{vis}} M_{\text{vis}}\|_F = 0, \quad \|M_{\text{vis}} T_{\partial}^{\text{vis}} - T_{\partial}^{\text{vis}} M_{\text{vis}}\|_F = 0, \tag{77}$$

which is the finite visible-block realization of the modular closure condition used in the full SHBT boundary partition function.

4 Boundary Entropy Densities and Self-Resolution

The modular data of Section 3 also determine how the finite visible register is loaded. The relevant coordinate lattice is the 3×3 visible block

$$\mathcal{C} = \{0, 1, 2\} \times \{0, 1, 2\}, \quad c = (i, j) \in \mathcal{C}. \tag{78}$$

The two axes of this lattice are the displayed $SU(2)_{26}$ charge slice and the displayed $SU(3)_8$ low-weight slice. Define

$$q_i, \ell_j \in (22, 23, 26), \quad w_i, w_j \in ((0, 0), (1, 0), (0, 1)), \tag{79}$$

where q_i and ℓ_j are the row and column labels of the same visible $SU(2)$ block. The raw boundary loading assigned to $c = (i, j)$ is the product of the modular overlap probabilities in the two visible factors,

$$\tilde{\rho}_B(i, j) = \left| S_{q_i, \ell_j}^{SU(2)_{26}} \right|^2 \left| S_{w_i, w_j}^{SU(3)_8} \right|^2. \tag{80}$$

This is positive on the benchmark visible block, so the finite normalization constant

$$Z_B^{\text{load}} = \sum_{(a, b) \in \mathcal{C}} \tilde{\rho}_B(a, b) \tag{81}$$

is nonzero. The normalized loading density is therefore

$$\rho_B(i, j) = \frac{\tilde{\rho}_B(i, j)}{\sum_{(a, b) \in \mathcal{C}} \tilde{\rho}_B(a, b)}, \quad \sum_{(i, j) \in \mathcal{C}} \rho_B(i, j) = 1. \quad (82)$$

The associated entanglement density is the normalized Shannon contribution of each loaded coordinate,

$$\rho_E(i, j) = \frac{-\rho_B(i, j) \ln \rho_B(i, j)}{\sum_{(a, b) \in \mathcal{C}} -\rho_B(a, b) \ln \rho_B(a, b)}, \quad \sum_{(i, j) \in \mathcal{C}} \rho_E(i, j) = 1. \quad (83)$$

The convention $0 \ln 0 = 0$ applies, although no zero entry occurs in the benchmark block.

The dominant loading sequence is the ordering of coordinates by decreasing normalized loading density,

$$\sigma = (c_1, \dots, c_9), \quad \rho_B(c_1) \geq \rho_B(c_2) \geq \dots \geq \rho_B(c_9). \quad (84)$$

Equal-density ties are physically degenerate and are resolved by a fixed lexicographic convention, which must agree with the coordinate order used by the entropy cascade. Equations (80)–(84) therefore define the following normalized objects:

- the loading density ρ_B , with $\sum_c \rho_B(c) = 1$,
- the entanglement density ρ_E , with $\sum_c \rho_E(c) = 1$, and
- the dominant sequence $\sigma = (c_1, \dots, c_9)$.

Entropy self-resolution promotes the static loading order into an iterative finite-register update. Let $U_n \subset \mathcal{C}$ be the unresolved set at step n , with $c_n \in U_n$ the next coordinate selected by σ . Before the update, the cumulative resolved entropy state is

$$\Omega_{n-1}(i, j) = \sum_{m < n} \delta_{c_m, (i, j)} \frac{\Delta S_m}{N}. \quad (85)$$

The resolved state feeds back into each unresolved coordinate through the Manhattan-distance kernel

$$F_{n-1}(c) = \sum_{(i, j) \in \mathcal{C}} \frac{\Omega_{n-1}(i, j)}{1 + |i - c_i| + |j - c_j|}, \quad c = (c_i, c_j). \quad (86)$$

For feedback coupling $\lambda \geq 0$, the loading weight is dressed as

$$w_B(c) = \rho_B(c)(1 + \lambda F_{n-1}(c)). \quad (87)$$

The entropy channel is dressed by the same feedback kernel,

$$w_E(c) = \rho_E(c)(1 + \lambda F_{n-1}(c)). \quad (88)$$

With remaining bit and entropy budgets

$$R_{n-1}^B = N - \sum_{m < n} \Delta B_m, \quad R_{n-1}^E = N - \sum_{m < n} \Delta S_m, \quad (89)$$

the self-resolution update is

$$\begin{aligned}\Delta B_n &= R_{n-1}^B \frac{w_B(c_n)}{\sum_{c \in U_n} w_B(c)}, \\ \Delta S_n &= R_{n-1}^E \frac{w_E(c_n)}{\sum_{c \in U_n} w_E(c)},\end{aligned}\tag{90}$$

$$\Delta T_n = \frac{\Delta S_n}{N}.\tag{91}$$

Thus ΔT_n is not an independent background clock tick. It is the dimensionless entropy residue left by resolving the next boundary coordinate. The terminal condition is fixed entirely by exhaustion of the normalized entropy budget.

The dimensional projection is driven by the cumulative temporal residue. At step n the effective projection dimension is

$$D_n = 26 - 22 \sum_{m \leq n} \Delta T_m, \quad \Delta D_n = D_{n-1} - D_n = 22 \Delta T_n.\tag{92}$$

Since U_9 contains only the final unresolved coordinate, the last update spends the full remaining entropy budget. By induction on Eq. (90),

$$\sum_{n=1}^9 \Delta S_n = N, \quad \sum_{n=1}^9 \Delta T_n = 1.\tag{93}$$

Consequently the benchmark projection starts at $D_0 = 26$ and terminates at

$$D_9 = 26 - 22 = 4.\tag{94}$$

This proves the $26 \rightarrow 4$ projection condition: it holds precisely when the self-resolution sequence exhausts the normalized entropy budget and reaches terminal dimension 4.

The same normalization gives the temporal kernel used for observer-frame cosmology. For a sampled expansion rate

$$H(t) = \frac{\dot{a}(t)}{a(t)},\tag{95}$$

the total entropy-production rate is postulated to saturate the finite register as

$$\dot{S}_{\text{total}} = N H(t).\tag{96}$$

The local entanglement contribution is distributed by the normalized density,

$$\dot{S}_{ij} = \rho_E(i, j) \dot{S}_{\text{total}}.\tag{97}$$

Therefore

$$\dot{T} = \frac{1}{N} \sum_{(i,j) \in \mathcal{C}} \dot{S}_{ij} = \frac{\dot{S}_{\text{total}}}{N} \sum_{(i,j) \in \mathcal{C}} \rho_E(i, j) = H(t).\tag{98}$$

This is the perception identity: apparent temporal flow is the entropy flow per finite boundary bit.

The full Section 4 closure structure is therefore

Physical object	Mathematical definition or closure relation
ρ_B	Normalized modular-overlap density, Eq. (82)
ρ_E	Normalized entanglement density, Eq. (83)
σ	Decreasing-loading order, Eq. (84)
F_{n-1} and $(\Delta B_n, \Delta S_n, \Delta T_n)$	Entropy self-resolution, Eq. (90)
$D_9 = 4$	Exhausted entropy budget, Eqs. (93)–(94)
$\dot{T} = H(t)$	Perception identity, Eq. (98)

The boundary entropy densities are therefore not phenomenological weights added after modular closure. They are the normalized modular-overlap probabilities of the finite visible block, and their self-resolution produces both the $26 \rightarrow 4$ dimensional projection and the local temporal kernel.

5 Holographic RG Flow: Boundary Entropy to Bulk Metric

The standard holographic starting point is the geometric dictionary for boundary entanglement. For a boundary subregion A , the Ryu–Takayanagi relation identifies its entanglement entropy with the area of the bulk minimal codimension-two surface γ_A homologous to A ,

$$S(A) = \frac{\text{Area}(\gamma_A)}{4G_{d+1}}. \quad (99)$$

This formula supplies the role of entanglement in the RG flow: changing the boundary resolution changes which extremal surfaces are sampled, and therefore which radial portion of the bulk geometry is probed [13, 10, 11, 12].

In an asymptotically anti-de Sitter geometry, the radial coordinate is the holographic renormalization scale. With the Fefferman–Graham form [6]

$$ds^2 = d\rho^2 + e^{2\rho/L} g_{ij}(x, \rho) dx^i dx^j = \frac{L^2}{z^2} (dz^2 + g_{ij}(x, z) dx^i dx^j), \quad (100)$$

the two common radial variables obey

$$z = L e^{-\rho/L}, \quad \frac{\mu(\rho)}{\mu_0} = e^{\rho/L} = \frac{L}{z}. \quad (101)$$

Thus moving outward toward the boundary corresponds to the ultraviolet of the dual CFT, while integrating out high-energy boundary modes corresponds to translating inward along the bulk radial direction. In the standard Hamilton–Jacobi formulation, the on-shell bulk action generates the boundary functional and radial evolution becomes a Callan–Symanzik flow,

$$\frac{\partial S_{\text{os}}}{\partial \rho} + \mathcal{H}_{\text{grav}} \left(\gamma_{ij}, \phi^I, \frac{\delta S_{\text{os}}}{\delta \gamma_{ij}}, \frac{\delta S_{\text{os}}}{\delta \phi^I} \right) = 0, \quad \mu \frac{dJ^I}{d\mu} = \beta^I(J). \quad (102)$$

This is the standard radial/Hamilton–Jacobi dictionary used in holographic renormalization [14, 15]. SHBT keeps the entanglement-to-geometry lesson of Eq. (99), but replaces the continuous large- N radial dynamics of Eq. (102) with a finite boundary-first projection. The input is the entropy

self-resolution cascade of Section 4; the output is a trace-normalized static metric slice. In this section $\tau \in [0, 1]$ denotes the normalized entropy-resolution parameter, not an independent external time coordinate. At the discrete resolution steps,

$$\tau_n = \sum_{m \leq n} \Delta T_m = \frac{1}{N} \sum_{m \leq n} \Delta S_m, \quad \tau_0 = 0, \quad \tau_9 = 1. \quad (103)$$

Thus the RG parameter is the same finite-register entropy residue that drives the $26 \rightarrow 4$ projection.

The finite SHBT replacement for the continuum boundary subregion is the visible coordinate lattice

$$\mathcal{C} = \{0, 1, 2\} \times \{0, 1, 2\}, \quad c = (i, j) \in \mathcal{C}. \quad (104)$$

The coordinates label the displayed $SU(2)_{26}$ charge slice and the $SU(3)_8$ low-weight slice. The raw boundary loading at $c = (i, j)$ is the product of the modular overlap probabilities in those two affine factors,

$$\tilde{\rho}_B(i, j) = \left| S_{q_i, \ell_j}^{SU(2)_{26}} \right|^2 \left| S_{w_i, w_j}^{SU(3)_8} \right|^2. \quad (105)$$

For the canonical branch, the explicit kernels used in this loading are

$$S_{ab}^{SU(2)_{26}} = \sqrt{\frac{2}{28}} \sin\left(\frac{\pi(a+1)(b+1)}{28}\right),$$

$$S_{(p,q),(r,s)}^{SU(3)_8} = \frac{i^3}{\sqrt{3} \, 11} \sum_{\sigma \in S_3} \text{sgn}(\sigma) \exp\left[-\frac{2\pi i}{11} (\sigma(\mathbf{v}(p, q) + \varrho), \mathbf{v}(r, s) + \varrho)\right], \quad (106)$$

where $\mathbf{v}(p, q)$ is the weight vector with Dynkin labels (p, q) and ϱ is the Weyl vector. Normalizing the raw loading gives

$$\rho_B(i, j) = \frac{\tilde{\rho}_B(i, j)}{\sum_{(a,b) \in \mathcal{C}} \tilde{\rho}_B(a, b)}, \quad \sum_{(i,j) \in \mathcal{C}} \rho_B(i, j) = 1. \quad (107)$$

The entropy density used by the RG projector is the normalized Shannon contribution of each loaded coordinate,

$$\rho_E(i, j) = \frac{-\rho_B(i, j) \ln \rho_B(i, j)}{\sum_{(a,b) \in \mathcal{C}} -\rho_B(a, b) \ln \rho_B(a, b)}, \quad \sum_{(i,j) \in \mathcal{C}} \rho_E(i, j) = 1. \quad (108)$$

Equations (104)–(108) are the Section 4 density definitions repeated here to make the RG derivation self-contained.

The prime lattice basis used by the metric projector is the five-point arithmetic skeleton

$$(p_0, p_1, p_2, p_3, p_4) = (2, 3, 5, 7, 11). \quad (109)$$

The choice of the prime basis $(2, 3, 5, 7, 11)$ is not a numerical coincidence or a compression trick. In information-theoretic terms, primes are the irreducible generators of the positive integers under multiplication; they represent the minimal set of atomic information channels required to encode the full arithmetic capacity of the finite register. The modulo-5 folding of the 9-component ordered entropy sequence onto this prime basis ensures that each residue class receives contributions from linearly independent arithmetic factors of the boundary load. This prevents spectral aliasing in the

subsequent Gram projection: if the folding were performed onto a composite basis, the Euler flux Φ_s would suffer from redundant factor cancellations that could wash out entropy shear. The prime basis therefore provides a maximal-entropy encoding of the ordered boundary state into a five-component load vector $\ell_r(\tau)$, where the logarithmic radial RG coordinates $x_r = \ln p_r$ are naturally ordered and have distinct prime-power scaling. This construction can be seen as a finite-register analogue of the continuous radial coordinate ρ in standard holographic RG, with the prime exponents playing the role of discrete energy scales. The five sites also match the rank of the parent D_5 algebra. If a discrete character weight is written in Dynkin coordinates as $\Lambda = (\lambda_1, \dots, \lambda_5)$, the conductor assignment $\mathfrak{c}(\Lambda) = 2^{\lambda_1} 3^{\lambda_2} 5^{\lambda_3} 7^{\lambda_4} 11^{\lambda_5}$ is injective by unique prime factorization. Consequently the skeleton gives an un-aliased prime-conductor decomposition of the five independent D_5 character-weight directions: distinct Dynkin weights cannot collapse to the same conductor, and every discrete parent-weight coordinate is represented. The specific set $(2, 3, 5, 7, 11)$ is the minimal ordered realization of this rank-five decomposition. Replacing it with any other five distinct ordered primes changes the numerical radial intervals but preserves the no-aliasing and closure structure.

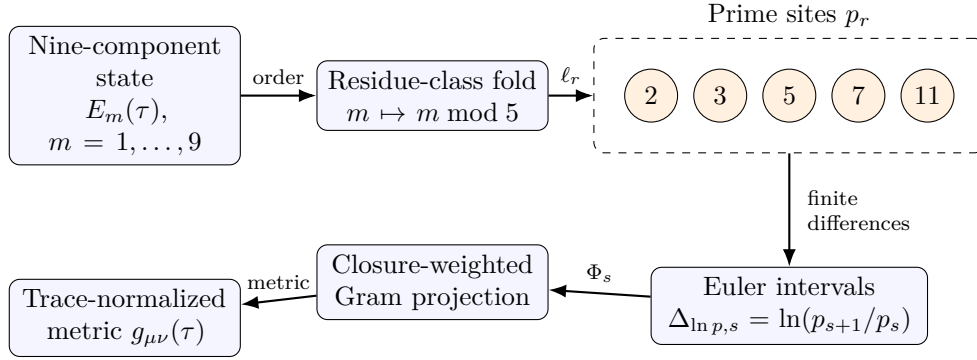


Figure 1: Prime-lattice RG folding. The ordered nine-component entropy state is folded by residue class onto the five prime sites; finite differences over the four logarithmic Euler intervals feed the closure-weighted Gram projection.

Its logarithms are the radial RG coordinates,

$$x_r = (\ln p_r), \quad r = 0, \dots, 4, \quad (110)$$

so nearest-neighbor finite differences on this lattice are naturally divided by $\ln(p_{s+1}/p_s) = x_{s+1} - x_s$. The role of the prime basis is not to introduce a new spectrum of particles. It is a fixed, finite coordinate frame in which the ordered boundary entropy state can be compressed into four Euler intervals and then lifted to a symmetric metric tensor.

Let

$$\sigma = (c_1, \dots, c_9) \quad (111)$$

be the dominant loading sequence of Eq. (84). The partially resolved entanglement state at RG parameter τ is the interpolation of the discrete self-resolution records,

$$\Omega_\tau(c) = \sum_{n=1}^9 \theta_n(\tau) \delta_{c_n, c} \Delta T_n, \quad (112)$$

where

$$\theta_n(\tau) = \begin{cases} 0, & \tau < \tau_{n-1}, \\ \frac{\tau - \tau_{n-1}}{\tau_n - \tau_{n-1}}, & \tau_{n-1} \leq \tau \leq \tau_n, \\ 1, & \tau > \tau_n. \end{cases} \quad (113)$$

Consequently $\sum_{c \in \mathcal{C}} \Omega_\tau(c) = \tau$. The metric projector uses both the static entanglement density and the resolved entropy state. Along the ordered sequence define

$$E_m(\tau) = \rho_E(c_m) + \Omega_\tau(c_m), \quad m = 1, \dots, 9. \quad (114)$$

Since $\sum_{m=1}^9 \rho_E(c_m) = 1$ and $\sum_{m=1}^9 \Omega_\tau(c_m) = \tau$, the total resolved load before folding is

$$\sum_{m=1}^9 E_m(\tau) = 1 + \tau. \quad (115)$$

The five-component load vector is obtained by folding this ordered sequence onto the prime lattice by residue class,

$$\begin{aligned} \hat{\ell}_r(\tau) &= \sum_{m=1}^9 E_m(\tau) \mathbf{1}_{\{m-1 \equiv r \pmod{5}\}}, \quad r = 0, \dots, 4, \\ \ell_r(\tau) &= \frac{\hat{\ell}_r(\tau)}{\sum_{q=0}^4 \hat{\ell}_q(\tau)} = \frac{\hat{\ell}_r(\tau)}{1 + \tau}. \end{aligned} \quad (116)$$

This gives

$$\ell_r(\tau) > 0, \quad \sum_{r=0}^4 \ell_r(\tau) = 1. \quad (117)$$

The dependence on σ is essential: the same multiset of boundary entropies would give a different prime-lattice load only if the modular loading order were changed. Equal-density ties retain the fixed lexicographic convention already specified in Section 4.

The Euler flux is the logarithmic finite difference of the load vector across adjacent prime sites. For $s = 0, 1, 2, 3$,

$$\Phi_s(\tau) = \frac{\ell_{s+1}(\tau) - \ell_s(\tau)}{\ln(p_{s+1}/p_s)}. \quad (118)$$

Thus Φ_s is the discrete RG gradient $\Delta\ell/\Delta \ln p$ on the finite prime lattice. The local mean loading and arithmetic hardness factor are

$$\bar{\ell}_s(\tau) = \frac{\ell_s(\tau) + \ell_{s+1}(\tau)}{2}, \quad h_s = \frac{1}{p_s} + \frac{1}{p_{s+1}}. \quad (119)$$

The closure-weighted contribution of the s th Euler interval is

$$W_s(\tau) = \lambda_{\text{closure}} (|\Phi_s(\tau)| + \bar{\ell}_s(\tau) + h_s), \quad \lambda_{\text{closure}} > 0. \quad (120)$$

Every term in Eq. (120) is nonnegative, so the outer-product part of the metric projection is positive semidefinite. The absolute flux measures local entropy shear, the mean load measures local occupancy, and h_s prevents a completely flat load vector from erasing the arithmetic interval.

The four Euler embedding vectors place the four intervals into the static metric block. Let $e_s^\mu = \delta_s^\mu$ for $\mu, s = 0, 1, 2, 3$, and read all load-vector indices modulo 5. The embedding vector is

$$v_s^\mu(\tau) = e_s^\mu + \left(\ell_s + \delta_{s0}, \ell_{s+1} + \delta_{s1}, \ell_{(s+2) \pmod 5} + \delta_{s2}, |\Phi_s| + \ell_{(s+3) \pmod 5} + \delta_{s3} \right)^\mu. \quad (121)$$

The Kronecker terms mark the Euler interval, while the cyclic load entries couple each interval to the neighboring prime sites. The fourth component also contains $|\Phi_s|$, so entropy shear enters the block as a metric direction rather than only as a scalar weight.

The unstabilized static metric is the weighted Gram matrix of these execution vectors, plus a positive diagonal closure floor,

$$\tilde{g}_{\mu\nu}(\tau) = \sum_{s=0}^3 W_s(\tau) v_{s\mu}(\tau) v_{s\nu}(\tau) + \text{diag}(d_0, d_1, d_2, d_3)_{\mu\nu}. \quad (122)$$

The diagonal entries are

$$d_s(\tau) = \lambda_{\text{closure}} + W_s(\tau) + \ell_s(\tau) + \ell_{s+1}(\tau), \quad s = 0, 1, 2, 3. \quad (123)$$

Since $\lambda_{\text{closure}} > 0$ and the remaining terms are nonnegative, Eq. (123) gives a strictly positive diagonal regularizer. The tensor in Eq. (122) is therefore already a finite, positive candidate for the static block; the final step enforces exact symmetry and trace normalization in the same way for every slice.

Define

$$A_{\mu\nu}(\tau) = \text{Sym}(\tilde{g}_{\mu\nu}(\tau) + \epsilon_{\text{eig}} \delta_{\mu\nu}), \quad \text{Sym}(B)_{\mu\nu} = \frac{B_{\mu\nu} + B_{\nu\mu}}{2}. \quad (124)$$

The eigenvalue lift is chosen as

$$\epsilon_{\text{eig}} = \max(0, \epsilon_{\text{pos}} - \lambda_{\min}(\text{Sym } \tilde{g})), \quad \epsilon_{\text{pos}} > 0. \quad (125)$$

In exact arithmetic the positive diagonal floor usually makes the lift unnecessary, but the construction retains Eq. (125) so that numerical roundoff cannot produce a nonpositive metric slice. The physical static metric is

$$g_{\mu\nu}(\tau) = \frac{A_{\mu\nu}(\tau)}{\text{Tr } A(\tau)} = \frac{\text{Sym}(\tilde{g}_{\mu\nu}(\tau) + \epsilon_{\text{eig}} \delta_{\mu\nu})}{\text{Tr}[\text{Sym } \tilde{g}(\tau) + \epsilon_{\text{eig}} \mathbf{1}_4]}. \quad (126)$$

It obeys the metric-slice checks

$$g_{\mu\nu} = g_{\nu\mu}, \quad \text{Tr } g = 1, \quad \lambda_{\min}(g) > 0. \quad (127)$$

The trace condition fixes the finite-register scale of the slice; it does not remove physical anisotropy, because anisotropy is carried by the normalized eigenvalue ratios and off-diagonal components inherited from the execution vectors.

The spatial bulk metric is the projection of the static 4×4 block onto the three rendered spatial directions. With

$$P_a^\mu = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \quad a = 1, 2, 3, \quad \mu = 0, 1, 2, 3, \quad (128)$$

the induced spatial metric is

$$g_{ab}^{\text{bulk}}(\tau) = P_a^\mu g_{\mu\nu}^{\text{static}}(\tau) P_b^\nu. \quad (129)$$

Equivalently, the spatial block is the lower-right 3×3 principal block of the stabilized static metric in this benchmark projector. Other projectors could rotate the rendered spatial frame, but the canonical projection uses the fixed matrix in Eq. (128) so that the benchmark is deterministic.

Combining the construction gives the complete holographic RG map

$$\boxed{\nabla_C S_\partial \mapsto \rho_E(i, j) \mapsto \ell_r(\tau) \mapsto \Phi_s(\tau), W_s(\tau), v_s^\mu(\tau) \mapsto g_{\mu\nu}(\tau) \mapsto g_{ab}^{\text{bulk}}(\tau)}. \quad (130)$$

The shortened form is

$$\nabla_C S_\partial \longrightarrow \rho_E(i, j) \longrightarrow \ell_r(\tau) \longrightarrow g_{\mu\nu}(\tau). \quad (131)$$

Every arrow is fixed by the previous sections: the boundary entropy gradient comes from modular-overlap loading, ρ_E is the normalized entanglement density, ℓ_r is the prime-lattice compression of the ordered resolved state, and $g_{\mu\nu}$ is the stabilized trace-one Gram projection. No independent bulk equation of motion is inserted between the boundary entropy state and the static metric slice.

This is the precise deviation from standard holographic RG. The usual AdS/CFT construction treats the bulk metric as a dynamical saddle governed by the Einstein–Hilbert action; the boundary scale dependence is recovered by solving radial Hamilton–Jacobi flow equations. SHBT instead uses the closed finite boundary register as the primitive object. The replacement is

$$(S(A), \rho, S_{\text{os}})_{\text{standard}} \longrightarrow (\rho_B, \rho_E, \Omega_\tau, x_r = \ln p_r)_{\text{SHBT}}, \quad \partial_\rho \longrightarrow \Delta_{\ln p}. \quad (132)$$

Thus Eq. (102) is not solved as an independent bulk field equation. Its SHBT analogue is the finite algebraic pipeline

$$(\rho_B, \rho_E, \Omega_\tau) \mapsto \ell_r(\tau) \mapsto \Phi_s(\tau) \mapsto \tilde{g}_{\mu\nu}(\tau) \mapsto g_{\mu\nu}(\tau), \quad (133)$$

with closure, positivity, and trace normalization checked slice by slice.

Feature	Standard holographic RG	SHBT RG flow
Primitive data	Continuum boundary QFT and classical bulk saddle	Completed finite boundary CFT and modular-overlap densities
Radial scale	Continuous ρ or z , with $\mu \propto e^{\rho/L} \propto z^{-1}$	Discrete prime sites $x_r = \ln p_r$ and Euler intervals $\Delta_{\ln p}$
Entanglement role	RT surfaces measure entropy from a given bulk metric	Normalized entropy densities synthesize the metric slice
Dynamics	Einstein–Hilbert equations recast as Hamilton–Jacobi flow	Deterministic entropy self-resolution and weighted Gram projection
Deviation	Large- N semiclassical bulk is assumed	No independent bulk dynamics is assumed; geometry is a closed projection

The physical closure map for this section is

Physical object	Mathematical definition or closure condition
Resolved entropy load	$\rho_E, \sigma, \Omega_\tau \mapsto \ell_r$
Discrete RG flux	$\Phi_s = \Delta \ell / \Delta \ln p$
Static metric synthesis	$\ell_r \mapsto \Phi_s, W_s, v_s^\mu, g_{\mu\nu}$
Spatial bulk projection	$P_a^\mu g_{\mu\nu} P_b^\nu$
Metric closure	Symmetry, unit trace, and positive eigenvalues
Slice data	$\ell_r, \Phi_s, g_{\mu\nu}, \gamma_{ab}$, and their spectra

The load construction in Eq. (116), the metric synthesis in Eqs. (118)–(126), and the spatial projection in Eq. (129) therefore form a single mathematical map. Its output is admissible only when Eq. (127) holds, the spatial block has dimension 3×3 , the slice order agrees with the dominant loading sequence, and no negative eigenvalue remains after stabilization.

6 Topological Baryogenesis and Anti-Baryon State Decoupling

The baryogenesis sector uses the completed boundary branch to separate two questions that are usually entangled. The first is the size of the visible baryon excess. The second is the fate of the anti-baryon stress-energy that would ordinarily accompany visible anti-baryon charge. The visible excess is fixed by a topological CP invariant and a modular restoration scale. The stress-energy is treated by an active Stinespring isometric state-projection map that strips visible color and electromagnetic charge while preserving the passive gravitational source. The mechanism is therefore not a dynamical erasure of stress-energy. It is a projection of active render channels into the dark completion sector, constrained by the same framing closure that fixes the branch.

The visible electroweak conversion factor is the standard sphaleron coefficient for the rendered Standard Model channel,

$$C_{\text{sph}} = \frac{28}{79}. \quad (134)$$

This coefficient converts the topological lepton-sector seed into the baryon number observed in the active visible register. In SHBT the CP-odd seed is not introduced as a free phase. It is the level-locked topological Jarlskog invariant

$$J_{\text{CP}}^{\text{topo}} = \frac{1}{K} \left(\frac{k_q}{k_\ell + 2} \right) \sqrt{1 - \kappa_{D_5}^2} \sin \left(\frac{2\pi k_q}{k_\ell} \right). \quad (135)$$

The factors have distinct origins. The parent completion contributes the dilution $1/K$; the visible level mismatch contributes $k_q/(k_\ell + 2)$, with the $+2$ the $SU(2)$ dual-Coxeter shift; the dark completion contributes the admissible overlap factor $\sqrt{1 - \kappa_{D_5}^2}$; and the modular phase difference between the quark and lepton clocks contributes $\sin(2\pi k_q/k_\ell)$. Thus CP violation is a residue of completed modular transport rather than an independently tunable angle.

The heavy restoration scale is fixed by the same branch arithmetic. Define the rank projection factor by

$$\Pi_{\text{rank}} = \sqrt{\frac{(h_{SO(10)}^\vee \dim SO(10))/12}{(h_{SU(3)}^\vee \dim SU(3))/12}} \sqrt{\frac{k_q + h_{SU(3)}^\vee}{K + h_{SO(10)}^\vee}}. \quad (136)$$

The residual matching of the 126 completion channel is written $\delta\Pi_{126}^{\text{match}}$. The modular restoration mass is then

$$M_N = M_{\text{GUT}} \exp\left[-\left(I_\ell^* \Pi_{\text{rank}} + I_q^* \delta\Pi_{126}^{\text{match}}\right)\right]. \quad (137)$$

For the canonical branch $I_\ell^* = 6$ and $I_q^* = 13$, so the exponent is a structural suppression determined by the completed embedding. It is not adjusted after the baryon asymmetry is computed.

The topological baryogenesis identity derived below is therefore a consistency check of the completed modular transport: the same algebra that fixes the visible levels and the dark completion residue also fixes the CP-odd seed and the restoration scale. The numerical outcome $\eta_B \approx 6.45 \times 10^{-10}$ is not a free parameter introduced to match observations; it is the residue of the modular phase mismatch between the quark and lepton clocks after the center lifts are locked. Thus the benchmark calculation reported in Section 8 is a verification that the branch arithmetic reproduces the observed visible asymmetry, but it does not claim to be an independent, unconstrained prediction of the baryon abundance without the branch data.

The SHBT baryogenesis identity is therefore

$$\eta_B = C_{\text{sph}} J_{\text{CP}}^{\text{topo}} \frac{M_N}{M_P}. \quad (138)$$

The four ingredients in Eq. (138) are fixed by the electroweak conversion coefficient, the modular CP residue, the restoration scale, and the Planck normalization. Substitution of the canonical branch data gives $\eta_B = 6.449923359416 \times 10^{-10}$.

The visible asymmetry becomes a complete accounting statement only after the anti-baryon channel is specified. Let $\mathcal{H}_B^{\text{vis}}$ denote the active visible anti-baryon Hilbert space. For a visible anti-baryon density matrix ρ , define the active render cost

$$\mathcal{C}_{\bar{B}}(\rho) = \gamma_3 \text{Tr}(\rho Q_{SU(3)}^2) + \gamma_{\text{EM}} \text{Tr}(\rho Q_{\text{EM}}^2) + \gamma_{\text{fr}} \Delta_{\text{fr}}^2, \quad \gamma_3, \gamma_{\text{EM}}, \gamma_{\text{fr}} > 0. \quad (139)$$

This is an active rendering functional, not a passive mass functional. It counts the cost of keeping anti-baryons visible as colored and electrically charged excitations, plus any residual framing mismatch. On the canonical branch $\Delta_{\text{fr}} = 0$, so the only nonzero active cost comes from visible gauge charge.

The Stinespring isometric state-projection operator is the completion-sector projection

$$\mathcal{D}_{\bar{B}} : \mathcal{H}_B^{\text{vis}} \longrightarrow \mathcal{H}_{\text{dark}}^{\text{B-L}}, \quad \rho_{\bar{B}}^{(n+1)} = \frac{\mathcal{D}_{\bar{B}} \rho_{\bar{B}}^{(n)} \mathcal{D}_{\bar{B}}^\dagger}{\text{Tr}(\mathcal{D}_{\bar{B}} \rho_{\bar{B}}^{(n)} \mathcal{D}_{\bar{B}}^\dagger)}. \quad (140)$$

The Stinespring isometric state-projection map $\mathcal{D}_{\bar{B}}$ acts at the modular restoration scale M_N given in Eq. (137). This scale is determined by the same branch arithmetic that fixes the parent completion and the framing defect. In the boundary theory, the transition is a continuous, unitary projection in the completed Hilbert space: as the energy scale descends from the GUT scale toward M_N , the visible anti-baryon gauge currents lose their support on the active $SU(3)$ and $U(1)_{\text{EM}}$ channels. Equivalently, the projection is a constant phase-transition boundary condition in the register: the active render cost $\mathcal{C}_{\bar{B}}$ drops to zero exactly at the scale where the modular restoration

exponent vanishes. For any finite Causal Point, however, this state decoupling is only observed as a local horizon-conditioned effect: if the observer's horizon fraction f_H (Eq. (145)) is too small, the active visible anti-baryon channels are already below the local entropy resolution limit, so the state decoupling appears to the local observer as a pre-existing completion asymmetry rather than a dynamical phase transition. Thus the mechanism has two complementary descriptions: (i) a bulk phase transition at M_N in the completed boundary data, and (ii) an observer-conditional projection for any Causal Point with f_H below the threshold required to resolve active anti-baryon gauge charges. The canonical branch fixes M_N once; the observer dependence only determines whether the local history record includes the transition, not whether the dark completion sector stores the passive stress-energy.

It is defined by the active charge-stripping constraints

$$\boxed{\mathcal{D}_{\bar{B}}^\dagger Q_{SU(3)} \mathcal{D}_{\bar{B}} = 0, \quad \mathcal{D}_{\bar{B}}^\dagger Q_{\text{EM}} \mathcal{D}_{\bar{B}} = 0.} \quad (141)$$

The dark target stores the conjugate $B - L$ residue, so the completed boundary state remains neutral even though the active visible anti-baryon gauge channels are no longer rendered.

Stress-energy preservation is imposed on the passive source, not on the active charge cost. The defining equality is

$$\boxed{\text{Tr}(\rho^{(n)} T_{\mu\nu}^{\bar{B}, \text{passive}}) = \text{Tr}(\rho^{(n+1)} T_{\mu\nu}^{\text{dark, passive}}).} \quad (142)$$

Thus the visible register can lose active anti-baryon color and electromagnetic support without violating the gravitational accounting used by the topological Einstein equation. Equivalently, the decoupled anti-baryon contributes as a dark passive source rather than as an actively rendered visible particle.

The anti-baryon loop closes at the fixed point

$$\boxed{\rho_{\bar{B}}^\star = \frac{\mathcal{D}_{\bar{B}} \rho_{\bar{B}}^\star \mathcal{D}_{\bar{B}}^\dagger}{\text{Tr}(\mathcal{D}_{\bar{B}} \rho_{\bar{B}}^\star \mathcal{D}_{\bar{B}}^\dagger)}, \quad \mathcal{C}_{\bar{B}}(\rho_{\bar{B}}^\star) = 0, \quad \delta T_{\mu\nu}^{\text{passive}} = 0.} \quad (143)$$

At this point the anti-baryon is absent from the active visible gauge-rendered channels, while its passive stress-energy has been moved into the dark completion sector. The observed baryon asymmetry is therefore a render-channel asymmetry, not an unbalanced disappearance of energy-momentum.

The physical closure conditions for this section are

Physical operator or observable	Mathematical definition or closure condition
Baryon asymmetry	$\eta_B = C_{\text{sph}} J_{\text{CP}}^{\text{topo}} M_N / M_P$
Active render cost	$\mathcal{C}_{\bar{B}}(\rho)$ in Eq. (139)
Charge-decoupling map	$\mathcal{D}_{\bar{B}}^\dagger Q_{SU(3)} \mathcal{D}_{\bar{B}} = \mathcal{D}_{\bar{B}}^\dagger Q_{\text{EM}} \mathcal{D}_{\bar{B}} = 0$
Passive source equality	Eq. (142)
Fixed-point state	$\rho_{\bar{B}}^\star$ with $\mathcal{C}_{\bar{B}} = 0$
Completed accounting	$\delta T_{\mu\nu}^{\text{passive}} = 0$

Together these conditions distinguish the active gauge-rendering functional from the passive gravitational source. The former vanishes at the projected fixed point, while the latter is exactly transferred to the dark completion sector. Implementation identifiers and performance diagnostics for this map are isolated in Section 12.

7 Causal Point and History Crystallization

The Causal Point is the observer interface of SHBT. It is not an external classical system appended to the boundary theory, but a finite local address inside the completed register. Its job is to translate global holographic closure into the quantities an observer can actually sample: a local memory budget, a local valuation of hidden information, a past-light-cone loading flow, and a deterministic projection that crystallizes one history from a finite outcome ensemble. Thus observation is treated as an entropy-budgeted operation on the boundary register rather than as a primitive background process.

The global horizon follows directly from the saturated bit budget introduced in the Introduction. Since $N = 3\pi/(L_P^2 \Lambda_{\text{holo}})$, the horizon radius is

$$R_H = \sqrt{\frac{NL_P^2}{\pi}} = \sqrt{\frac{3}{\Lambda_{\text{holo}}}}. \quad (144)$$

For a Causal Point whose local sampling radius is R_{local} , define the horizon fraction and local area by

$$f_H = \frac{R_{\text{local}}}{R_H}, \quad A_{\text{local}} = 4\pi R_{\text{local}}^2, \quad 0 \leq f_H \leq 1. \quad (145)$$

The observer cannot address the full horizon register unless $f_H = 1$. The available local memory, hidden complement, and operational entropy limit are therefore

$$N_{\text{local}} = N f_H^2, \quad N_{\text{hidden}} = N - N_{\text{local}}, \quad N_{\text{limit}} = \min\left(N_{\text{local}}, \frac{A_{\text{local}}}{4L_P^2 \ln 2}\right). \quad (146)$$

The first entry counts addressable register bits. The second counts degrees of freedom outside the local addressable patch. The third enforces the stricter of the local register count and the local area-law bound, with the factor $\ln 2$ converting the area entropy into bits.

The Causal Point's self-valuation is the scalar response of the local observer to residual framing and hidden-sector loading. Writing $\Delta_{\text{frame}} \equiv \Delta_{\text{fr}}$ and

$$f_{\text{hidden}} = \frac{N_{\text{hidden}}}{N} = 1 - \frac{N_{\text{local}}}{N}, \quad w(\xi) = \frac{\xi_\ell + \xi_q + \xi_s}{3}, \quad (147)$$

the valuation field is

$$\Sigma = (1 + \Delta_{\text{frame}})(1 + w(\xi)f_{\text{hidden}}). \quad (148)$$

On the canonical branch $\Delta_{\text{frame}} = 0$, but the formula is written in a branch-independent form so that any residual framing mismatch would be visible as a multiplicative observer cost. The localized

entropy gradient is the valuation density across the local sampling radius,

$$\nabla_{\text{obs}} \Sigma = \frac{\Sigma f_{\text{hidden}}}{R_{\text{local}}}. \quad (149)$$

The apparent observer acceleration is then

$$a_{\text{obs}} = c^2 \nabla_{\text{obs}} \Sigma. \quad (150)$$

This acceleration is not inserted as an external force law. It is the local inertial response to the hidden entropy gradient seen by an entropy-limited observer.

The same local data define the observer-frame Jacobian. The anisotropy coordinates (ξ_ℓ, ξ_q, ξ_s) are the three directional entries stored in the local property packet, and the Causal Point projection matrix is

$$J^\mu{}_\nu = \begin{pmatrix} 1 + \Sigma f_{\text{hidden}} & -\Sigma \xi_\ell & -\Sigma \xi_q & -\Sigma \xi_s \\ 0 & 1 + \Sigma \xi_\ell & 0 & 0 \\ 0 & 0 & 1 + \Sigma \xi_q & 0 \\ 0 & 0 & 0 & 1 + \Sigma \xi_s \end{pmatrix}. \quad (151)$$

Applied to a bulk metric slice from Section 5, it gives the observer-frame metric

$$g_{\mu\nu}^{\text{obs}} = J_\mu{}^\alpha g_{\alpha\beta} J_\nu{}^\beta. \quad (152)$$

The off-diagonal entries in the first row encode the finite-register mixing between temporal sampling and the three anisotropic property channels, while the diagonal spatial entries encode direction-dependent local rescaling.

Past-light-cone flow is the observer-time image of a static boundary loading process. Let $N_{\text{load}}(t)$ be the number of loaded bits along the sampled history. The loading fraction and effective expansion rate are

$$f_{\text{load}}(t) = \frac{N_{\text{load}}(t)}{N}, \quad H_{\text{eff}}(t) = H_\Lambda + \frac{1}{3} \frac{df_{\text{load}}}{dt}. \quad (153)$$

Here H_Λ is the horizon rate associated with the static Λ_{holo} sector. The second term is not a new external clock; it is the one-dimensional residue of finite boundary loading as reconstructed by the Causal Point. Along the observer's past light cone, redshift parameterizes the same process through

$$\frac{dt}{dz} = -\frac{1}{(1+z)H_{\text{eff}}(z)}, \quad \frac{d\chi}{dz} = \frac{1}{H_{\text{eff}}(z)}. \quad (154)$$

The first equation gives the lookback-time increment, and the second gives the comoving radial increment in the same light-cone units. A sampled redshift therefore selects a boundary coordinate by the ordered loading sequence,

$$n(z) = 1 + \lfloor (|\sigma| - 1) f_{\text{load}}(z) \rfloor, \quad c(z) = \sigma_{n(z)} \in \mathcal{C}. \quad (155)$$

This gives the bridge between the apparent cosmological flow and the finite boundary address on which a measurement can be attempted.

History crystallization begins only after such a boundary address has been selected. Let $\mathcal{R}$ be the finite register visible to the Causal Point, let $A \in \mathcal{R}$ be the selected address, and let

$$\Omega_A = \{\omega_0, \omega_1, \dots, \omega_{m-1}\}, \quad \sum_{i=0}^{m-1} p_i = 1, \quad p_i \geq 0, \quad (156)$$

be the finite outcome ensemble and its normalized local loading weights. The entropy costs of a retrieval event are

$$\begin{aligned} C_{\text{addr}} &= \log_2 |\mathcal{R}|, & C_{\text{ens}} &= \log_2 |\Omega_A|, \\ C_{\text{get}} &= \max(1, C_{\text{addr}} + C_{\text{ens}}, C_{\text{req}}), & R_{\text{entropy}} &= N_{\text{limit}} - C_{\text{get}}. \end{aligned} \quad (157)$$

The GET operation is admissible exactly when

$$R_{\text{entropy}} \geq 0. \quad (158)$$

If the residual is negative, the Causal Point has insufficient local entropy budget to crystallize the requested classical record.

When the budget condition holds, the deterministic collapse index is

$$\boxed{\iota = \text{Hash}(\mathcal{O}, A, f_H, N_{\text{local}}, m) \bmod m.} \quad (159)$$

The corresponding rank-one projection operator is

$$\boxed{\Pi_{A,\iota} = |A, \omega_\iota\rangle\langle A, \omega_\iota|.} \quad (160)$$

For a sequence of admissible Causal Point retrievals, the history density matrix updates by

$$\rho_{\text{hist}}^{(n+1)} = \frac{\Pi_{A_n, \iota_n} \rho_{\text{hist}}^{(n)} \Pi_{A_n, \iota_n}}{\text{Tr}(\Pi_{A_n, \iota_n} \rho_{\text{hist}}^{(n)})}. \quad (161)$$

The full crystallized history along a sampled past light cone is therefore the ordered product

$$\begin{aligned} \Pi_{\text{history}}(z_1, \dots, z_m) &= \Pi_{A_m, \iota_m} \cdots \Pi_{A_2, \iota_2} \Pi_{A_1, \iota_1}, \\ A_n &= A(c(z_n)). \end{aligned} \quad (162)$$

Thus a classical past is not retrieved from an unlimited archive. It is the finite ordered record that survives all local entropy-budget checks.

The pointer wavefunction associated with the selected outcome is the visible and dark completion packet

$$\boxed{\Psi_\iota = (A_\iota = p_\iota, \quad c_{\text{vis}} = -p_\iota, \quad c_{\text{dark}} = p_\iota).} \quad (163)$$

The opposite signs of the visible and dark coefficients encode the same completion accounting used elsewhere in SHBT: the visible pointer is rendered as a local classical record while the complementary dark coefficient preserves the finite-register balance.

The physical closure structure for this section is

Physical object	Mathematical definition or closure condition
Causal observer address	$(A, f_H, N_{\text{local}})$
Past-light-cone sample	Eqs. (153)–(155)
Local property packet	Self-valuation and observer Jacobian
Entropy-budget admissibility	$N_{\text{hidden}} + C_A \leq N_{\text{local}}$
History projection	$\Pi_{A,\ell}$ and Π_{history}

These relations combine the local horizon fraction, hidden-bit fraction, self-valuation, observer Jacobian, and past-light-cone samples into a single admissibility problem. Equations (146) and (158) determine whether the history operators in Eqs. (159)–(163) may act. The software records implementing these quantities are documented in Section 12.

8 Numerical Verification of Closure

All checks in this section are closure tests, not phenomenological fits. The branch constants, central charges, and finite register size are fixed by the modular closure axioms of Sections 2–3; the numerical layer confirms that these axioms close consistently across the entropy, metric, baryogenesis, and observer sectors. Where a derived value agrees with a measured cosmological quantity (e.g., η_B), that agreement is read as evidence that the completed boundary data are physically relevant, not as a validation of a free parameter.

The preceding sections define SHBT by algebraic closure, entropy self-resolution, holographic projection, Causal Point admissibility, and topological anti-baryon state decoupling. This section records the corresponding finite-dimensional physical outputs. Software entry points and reproducibility commands are collected separately in Section 12. The verification is not a separate phenomenological fit: every entry is evaluated on the same canonical branch

$$(k_\ell, k_q, K) = (26, 8, 312), \quad \Delta_{\text{fr}} = 0. \quad (164)$$

The purpose of the numerical layer is therefore to check that the branch data, normalization conventions, projector algebra, observer memory accounting, and baryogenesis identities all close simultaneously.

8.1 Boundary closure verification

The canonical boundary is tested before any bulk metric or observer-frame reconstruction is used. The first requirement is exact framing closure:

$$\Delta_{\text{fr}}(312, 26, 8) = \max\left(\left\|\frac{312}{2 \cdot 26}\right\|_{\mathbb{Z}}, \left\|\frac{312}{3 \cdot 8}\right\|_{\mathbb{Z}}\right) = 0, \quad (165)$$

The loading and entanglement densities used by the entropy flow are normalized. The modular tests verify that the visible and completed partition data commute with the constructed S_∂ and T_∂ kernels:

$$\|[M, S_\partial]\| = 0.0, \quad \|[M, T_\partial]\| = 0.0. \quad (166)$$

The full boundary closure ledger is summarized in Table 2.

Physical operator or observable	Mathematical definition	Theoretical value	Closure condition
Affine branch	(k_ℓ, k_q, K)	$(26, 8, 312)$	$I_\ell^* = 6, I_q^* = 13$
Framing defect	$\Delta_{\text{fr}}(K, k_\ell, k_q)$	0	exact center-lift closure
$Z_\partial^{\text{vis}}(i)$	$\chi^\dagger M_{\text{vis}} \chi$	$2.441381789163 \times 10^{-24}$	modular-orbit equality
Loading density	$\sum_c \rho_B(c)$	1	normalized
Entanglement density	$\sum_c \rho_E(c)$	1	normalized
Dominant sequence	$ \sigma = \mathcal{C} $	9	complete ordering
Modular S closure	$\ [M, S_\partial]\ _F$	0	exact commutation
Modular T closure	$\ [M, T_\partial]\ _F$	0	exact commutation
Boundary Hamiltonian	$\Pi_{\text{phys}} H_\partial \Pi_{\text{phys}}$	0	zero-energy lock
Visible dimension	D_9	4	exhausted entropy budget

Table 2: Physical boundary observables and closure conditions on the canonical branch.

Together these checks establish the finite-dimensional form of the first closure equivalence:

$$[M, S_\partial] = [M, T_\partial] = 0 \iff \Delta_{\text{fr}} = 0 \implies \Pi_{\text{phys}} H_\partial \Pi_{\text{phys}} = 0. \quad (167)$$

The final two entries in Table 2 show that the completed static boundary has no residual external time generator and that the visible lepton-sector projection reaches the expected terminal dimension.

8.2 Bulk metric projection

The normalized loading and entanglement densities determine the entropy cascade of metric slices. The canonical branch gives the full nine-slice cascade,

$$n_{\text{slice}} = 9, \quad (168)$$

with a rank-three spatial projector at every sampled step. Each metric slice is then checked for the three algebraic properties required by the projection construction:

$$g_{ij} = g_{ji}, \quad \text{Tr}(g) = 1, \quad v^i g_{ij} v^j > 0 \quad \text{for } v \neq 0. \quad (169)$$

The resulting closure table is Table 3.

Physical operator or observable	Mathematical definition	Theoretical value	Closure condition
Entropy cascade	$n_{\text{slice}} = \mathcal{C} $	9	complete resolution
Spatial projector	$\text{rank } P$	3	three-dimensional image
Metric symmetry	$g_{ij} - g_{ji}$	0	symmetric
Trace normalization	$\text{Tr } g$	1	normalized
Metric spectrum	$\lambda_{\min}(g)$	> 0	positive definite

Table 3: Physical metric observables and closure conditions for the holographic projection.

The entropy flow therefore produces symmetric, trace-normalized, positive metric data with the intended rank-three spatial image. The bulk slices thus satisfy the finite-dimensional projection conditions stated in Section 5.

8.3 Causal Point closure

Observer access is limited by a finite entropy budget. For a horizon fraction f_H , the local bit budget satisfies

$$N_{\text{local}}^{\text{obs}} \simeq N f_H^2, \quad N_{\text{hidden}}^{\text{obs}} = N - N_{\text{local}}^{\text{obs}}. \quad (170)$$

The retrieval interface is admissible only when the entropy limit is positive, and the past-light-cone sample count must match the chosen redshift grid:

$$N_{\text{limit}}^{\text{obs}} > 0, \quad n_{\text{lightcone}}^{\text{obs}} = n_{\text{redshift}}. \quad (171)$$

The closure values are summarized in Table 4.

Physical operator or observable	Mathematical definition	Theoretical value	Closure condition
Local memory	$N_{\text{local}}^{\text{obs}}$	$2.535748254483999 \times 10^{122}$	finite
Hidden register	$N_{\text{hidden}}^{\text{obs}}$	$7.76249465658367 \times 10^{121}$	finite
Entropy limit	$N_{\text{limit}}^{\text{obs}}$	$2.535748254483999 \times 10^{122}$	positive
Light-cone samples	$n_{\text{lightcone}}^{\text{obs}}$	9	redshift-grid matched
Property packets	n_{packet}	9	one per sample
Memory inequality	$N_{\text{hidden}} + C_A \leq N_{\text{local}}$	satisfied	admissible

Table 4: Causal Point observables and finite-memory closure conditions.

These values check the observer side of the closure chain. The Causal Point does not access an unlimited record; it constructs exactly the number of past-light cone entries required by the redshift sampler, constructs nine local property packets, and filters each attempted retrieval through the finite-memory inequality.

8.4 Baryogenesis and projected field accounting

The baryogenesis comparison contrasts an unprojected visible field description with the SGBT fixed point in which anti-baryon visible charge channels are decoupled while their passive stress-energy contribution remains in the dark completion sector. The visible baryon asymmetry is

$$\eta_B = 6.449923359416 \times 10^{-10}, \quad (172)$$

with exact preservation of the passive stress-energy source. The physical state-space comparison is shown in Table 5.

The physical comparison is shown in Table 6. The projected description is not allowed to discard energy: it removes anti-baryon charge from the actively rendered visible channel but preserves the passive source that enters the topological Einstein equation.

The combined verification therefore establishes modular invariance, the zero-energy lock, finite-memory admissibility, and passive stress-energy preservation on the same canonical branch.

Field description	Baryon support	Anti-baryon support	Passive gravitational source	Closure
Unprojected visible space	active	active	visible-sector source	baseline
SHBT projected fixed point	active	gauge-decoupled	dark-completion source	$\delta T_{\mu\nu}^{\text{passive}} = 0$
Render-channel change	unchanged	removed from active gauge channels	exactly preserved	closed

Table 5: Physical state-space comparison before and after anti-baryon gauge-channel decoupling.

Physical quantity	Unprojected visible description	SHBT projected description
Visible baryon channel	actively rendered	actively rendered
Visible anti-baryon channel	actively rendered	decoupled
Passive stress-energy	explicitly evolved	preserved in dark completion
Baryon asymmetry output	dynamical baseline target	$\eta_B = 6.449923359416 \times 10^{-10}$
Stress-energy check	baseline conservation test	true

Table 6: Physical accounting in the unprojected and SHBT-projected field descriptions.

9 Precision Cosmological Implications

The static closed boundary theory induces an observer-frame cosmology for a finite observer within the register. No additional branch coordinate is introduced. The center-locked parent level, dark completion, finite entropy register, and Causal Point light-cone rule determine the corresponding precision-cosmology effective description. The derivation is independent of the implementation interface; the exact function and report-key mapping is given in Section 12.

9.1 Completion Notation and Entropy Debt

The canonical branch requires distinct residual and completed dark ledgers. Their definitions, together with the entropy-debt variable, are given by

$$c_{\text{dark}}^{\text{res}} = \frac{834433}{362670}, \quad c_{\text{dark}}^{\text{comp}} = c_{\text{dark}}^{\text{res}} + 1 = \frac{1197103}{362670}, \quad \Delta_{\text{mod}} = \frac{c_{\text{dark}}^{\text{comp}}}{24} = \frac{1197103}{8704080}. \quad (173)$$

The residual quantity $c_{\text{dark}}^{\text{res}}$ is the modular closure residue, whereas $c_{\text{dark}}^{\text{comp}}$ is the finite-screen completion entering the Noether bridge and the entropy debt. Appendix B.4, Eq. (295), gives the completed ledger from the coset subtraction, while Appendix A fixes the parent-level arithmetic used by both ledgers. These definitions are used throughout.

The finite horizon register is the saturated SHBT form of the holographic entropy bound. For a spatial region bounded by a cosmological horizon of area A , the Bekenstein–Hawking counting bound is

$$S \leq \frac{A}{4L_P^2 \ln 2}. \quad (174)$$

On the canonical branch the cosmological screen is fixed by Λ_{holo} , so the saturated bit capacity is

$$N_{\text{sat}} = \frac{3\pi}{L_P^2 \Lambda_{\text{holo}}}, \quad \Lambda_{\text{holo}} = 1.08913883 \times 10^{-52} \text{ m}^{-2}, \quad N_{\text{sat}} = 3.312593327986 \times 10^{122}. \quad (175)$$

This finite number is not a density parameter. It is the total horizon register against which the observer-frame loading process is charged.

Let $N_{\text{load}}(z)$ be the number of horizon bits actively loaded by the past-light-cone sample at redshift z . The dimensionless loading fraction and entropy debt are

$$f_{\text{load}}(z) = \frac{N_{\text{load}}(z)}{N_{\text{sat}}}, \quad S_{\text{debt}}(z) = N_{\text{load}}(z) = N_{\text{sat}} f_{\text{load}}(z), \quad 0 \leq f_{\text{load}}(z) \leq 1. \quad (176)$$

The horizon contribution is saturated on this branch, so the only active light-cone entropy increment is the debt term. If t_{lc} denotes the observer's past-light-cone acquisition parameter, then

$$\frac{dz}{dt_{\text{lc}}} = (1+z)H_{\text{SHBT}}(z). \quad (177)$$

The Causal Point construction imposes a constant loading rate Γ_{lock} in the clock-normalized channel:

$$\frac{1}{N_{\text{sat}}} \frac{dS_{\text{total}}}{dt_{\text{lc}}} = \frac{\Gamma_{\text{lock}}}{1+z}. \quad (178)$$

Combining Eqs. (176)–(178) gives

$$\begin{aligned} \frac{dS_{\text{total}}}{dt_{\text{lc}}} &= \frac{dS_{\text{H}}}{dt_{\text{lc}}} + \frac{dS_{\text{debt}}}{dt_{\text{lc}}} = 0 + N_{\text{sat}} \frac{df_{\text{load}}}{dz} \frac{dz}{dt_{\text{lc}}} \\ &= N_{\text{sat}} \frac{df_{\text{load}}}{dz} (1+z)H_{\text{SHBT}}(z) = N_{\text{sat}} \frac{\Gamma_{\text{lock}}}{1+z} > 0. \end{aligned} \quad (179)$$

Solving the last equality for the loading fraction gives the redshift ODE

$$\boxed{\frac{df_{\text{load}}}{dz} = \frac{\Gamma_{\text{lock}}}{(1+z)^2 H_{\text{SHBT}}(z)}}. \quad (180)$$

Thus the generalized second law is not an independent late-time fluid assumption; it is the monotone light-cone image of finite-register loading.

The lock rate is fixed by the Causal Point clock relation

$$d\tau_{\text{lock}} = \frac{dt}{1+z}. \quad (181)$$

In the observer frame, the loaded clock channel contributes one third of its rate to the locally inferred Hubble intercept:

$$H_0(z) = H_{\Lambda} + \frac{1}{3(1+z)} \frac{df_{\text{load}}}{d\tau_{\text{lock}}}. \quad (182)$$

The static horizon term is anchored to the CMB intercept, $H_{\Lambda} = H_0^{\text{CMB}}$. Evaluating Eq. (182) at $z = 0$ gives

$$H_0^{\text{loc}} = H_0^{\text{CMB}} + \frac{1}{3} \frac{df_{\text{load}}}{d\tau_{\text{lock}}}, \quad \Gamma_{\text{lock}} \equiv \frac{df_{\text{load}}}{d\tau_{\text{lock}}} = 3(H_0^{\text{loc}} - H_0^{\text{CMB}}). \quad (183)$$

This identifies the entropy-debt loading rate with the measured separation between the local and CMB Hubble intercepts.

9.2 Noether Bridge and Newton Lock

The Noether bridge construction uses the completed ledger to evaluate

$$8\pi G_{\text{eff}} = \frac{12}{c_{\text{dark}}^{\text{comp}} M_P^2}, \quad \frac{d \ln G_{\text{eff}}}{dt} = -\frac{d \ln c_{\text{dark}}^{\text{comp}}}{dt} - 2 \frac{d \ln M_P}{dt}. \quad (184)$$

On the fixed branch both logarithmic derivatives vanish, so the Bianchi stationarity check gives $\dot{G}/G = 0$. The same branch verifies the unity-of-scale identity. The retained D_5 cell factor κ_{D_5} entering this identity is defined in Appendix B, Eq. (292), and the neutrino eigenvalues used for the normal hierarchy are listed in Eqs. (293)–(294):

$$m_{\nu,1} = \kappa_{D_5} M_P N^{-1/4}, \quad \Lambda_{\text{holo}} = \frac{3\pi}{\kappa_{D_5}^4} G_{\text{top}} m_{\nu,1}^4, \quad \epsilon_\Lambda \leq \frac{1}{N}. \quad (185)$$

The bridge quantities are collected in Table 7. The Newton-lock identities above remain analytic fixed-branch consequences.

Physical operator or observable	Mathematical definition	Theoretical value	Closure condition
Completed ledger	$c_{\text{dark}}^{\text{comp}} = c_{\text{dark}}^{\text{res}} + 1$	$1197103/362670 = 3.300805139659$	exact rational closure
Entropy debt D_5 cell measure	$\Delta_{\text{mod}} = c_{\text{dark}}^{\text{comp}}/24$ κ_{D_5}	0.137533547486 0.988769793998	fixed by ledger retained boundary measure
Holographic curvature	Λ_{holo}	$1.08913883 \times 10^{-52} \text{ m}^{-2}$	fixed screen
Horizon register	$N(\Lambda_{\text{holo}}) = 3\pi/(L_P^2 \Lambda_{\text{holo}})$	$3.312593327986 \times 10^{122}$	finite saturation
CMB intercept	H_0^{CMB}	$67.4 \text{ km s}^{-1} \text{ Mpc}^{-1}$	early-time anchor
Neutrino floor	$m_{\nu,1} = \kappa_{D_5} M_P N^{-1/4}$	$2.829630635353 \times 10^{-3} \text{ eV}$	unity of scale
Normal mass sum	Σ_ν^{NH}	61.8 meV	lower-tension branch
Inverted mass sum	Σ_ν^{IH}	103 meV	comparison branch

Table 7: Completed-ledger observables, theoretical values, and Noether-bridge closure conditions.

9.3 Neutrino Hierarchy Lock

The same completed cosmological closure fixes the neutrino floor before the Hubble loading law is applied. The lightest normal eigenstate is not a fit parameter: it is the finite-register quarter-root scale weighted by the retained D_5 boundary-cell measure,

$$\boxed{m_{\nu,1} = \kappa_{D_5} M_P N_{\text{sat}}^{-1/4}}. \quad (186)$$

Substituting the canonical values gives

$$\begin{aligned} m_{\nu,1} &= 0.988769793998 \left(1.220890146939 \times 10^{28} \text{ eV} \right) \left(3.312593327986 \times 10^{122} \right)^{-1/4} \\ &= 2.829630635353 \times 10^{-3} \text{ eV}. \end{aligned} \quad (187)$$

The oscillation splittings then convert the boundary floor into absolute propagation eigenstates:

$$\Delta m_{21}^2 \simeq 7.4 \times 10^{-5} \text{ eV}^2, \quad \Delta m_{31}^2 \simeq 2.5 \times 10^{-3} \text{ eV}^2. \quad (188)$$

For the normal hierarchy,

$$m_{\nu,2}^{\text{NH}} = \sqrt{m_{\nu,1}^2 + \Delta m_{21}^2} \simeq 8.92 \times 10^{-3} \text{ eV}, \quad m_{\nu,3}^{\text{NH}} = \sqrt{m_{\nu,1}^2 + \Delta m_{31}^2} \simeq 5.01 \times 10^{-2} \text{ eV}. \quad (189)$$

The normal-ordering sum is therefore

$$\Sigma_{\nu}^{\text{NH}} = m_{\nu,1}^{\text{NH}} + m_{\nu,2}^{\text{NH}} + m_{\nu,3}^{\text{NH}} = 61.8 \text{ meV}. \quad (190)$$

For the inverted hierarchy, the finite-register floor is assigned to the lightest third eigenstate,

$$m_{\nu,3}^{\text{IH}} = 2.829630635353 \times 10^{-3} \text{ eV}, \quad m_{\nu,1}^{\text{IH}} \simeq 4.87 \times 10^{-2} \text{ eV}, \quad m_{\nu,2}^{\text{IH}} \simeq 4.95 \times 10^{-2} \text{ eV}, \quad (191)$$

so that

$$\Sigma_{\nu}^{\text{IH}} = m_{\nu,3}^{\text{IH}} + \sqrt{(m_{\nu,3}^{\text{IH}})^2 + |\Delta m_{32}^2|} + \sqrt{(m_{\nu,3}^{\text{IH}})^2 + |\Delta m_{32}^2| + \Delta m_{21}^2} = 103 \text{ meV}. \quad (192)$$

The hierarchy statistic is the normalized pull of each derived sum against the cosmological mass-sum likelihood:

$$\sigma_{\text{tens}}^{\text{NH}} = \frac{\Delta \Sigma_{\nu}^{\text{NH}}}{\sigma_{\Sigma}^{\text{NH}}} = 1.68, \quad \sigma_{\text{tens}}^{\text{IH}} = \frac{\Delta \Sigma_{\nu}^{\text{IH}}}{\sigma_{\Sigma}^{\text{IH}}} = 2.80. \quad (193)$$

Thus the normal ordering is the lower-tension finite-register completion. The explicit eigenstate ledger is shown in Table 8.

Physical observable	Mathematical definition	Normal hierarchy	Inverted hierarchy
Floor eigenstate	$\kappa_{D_5} M_P N_{\text{sat}}^{-1/4}$	$m_1 = 2.83 \times 10^{-3} \text{ eV}$	$m_3 = 2.83 \times 10^{-3} \text{ eV}$
Second eigenstate	oscillation-splitting lift	$m_2 = 8.92 \times 10^{-3} \text{ eV}$	$m_1 = 4.87 \times 10^{-2} \text{ eV}$
Third eigenstate	oscillation-splitting lift	$m_3 = 5.01 \times 10^{-2} \text{ eV}$	$m_2 = 4.95 \times 10^{-2} \text{ eV}$
Mass sum	$\Sigma_{\nu} = \sum_i m_{\nu,i}$	61.8 meV	103 meV
Likelihood pull	$\Delta \Sigma_{\nu} / \sigma_{\Sigma}$	1.68 σ	2.80 σ

Table 8: Neutrino propagation eigenstates, mass sums, and hierarchy likelihood pulls.

The hierarchy check is therefore tied to the same κ_{D_5} , N_{sat} , and completed-ledger inputs as the Newton lock.

9.4 Cosmological Observable Map

The precision sector is organized by the physical observables and closure relations in Table 9:

Table 9 is the cross-reference index for the precision-cosmology derivations: the completed ledger and finite screen are Eqs. (173)–(175); the light-cone loading and GSL identities are Eqs. (176)–(180); the Noether bridge and neutrino hierarchy are Eqs. (184)–(193); the observer-frame Hubble law is Eqs. (194)–(199); growth, mirage, cluster-collapse, and dark-matter outputs are tied to Eqs. (200)–(206) and Tables 11–13, with the dark-matter ledger in Eqs. (DM1)–(DM3) and Table 19; ISW, BBN, and chronometer checks are Eqs. (207)–(221); and the GET measurement rule is Eqs. (222)–(236).

Physical operator or observable	Mathematical definition	Theoretical role or value	Closure condition
Completed ledger	$c_{\text{dark}}^{\text{comp}} = c_{\text{dark}}^{\text{res}} + 1$	finite-screen dark contribution	exact rational closure
D_5 measure	κ_{D_5}	retained boundary-cell weight	quantum-dimension closure
Horizon register	$N_{\text{sat}} = 3\pi/(L_P^2 \Lambda_{\text{holo}})$	finite information capacity	saturated screen
Neutrino floor	$m_{\nu,1} = \kappa_{D_5} M_P N_{\text{sat}}^{-1/4}$	absolute mass scale	unity of scale
Entropy debt	$\Delta_{\text{mod}} = c_{\text{dark}}^{\text{comp}}/24$	Hubble uplift parameter	fixed by ledger
Local intercept	$H_0^{\text{loc}} = H_0^{\text{CMB}} e^{\Delta_{\text{mod}}/2}$	late-time Hubble endpoint	CMB anchored
Loading rates	$A_H = H_0^{\text{loc}} - H_0^{\text{CMB}}, \Gamma_{\text{lock}} = 3A_H$	amplitude and lock rate	causal-clock lock
Hubble law	$H_0(z) = H_0^{\text{CMB}} + A_H/(1+z)$	observer-frame interpolation	correct endpoints
Entropy loading	$f_{\text{load}}, S_{\text{debt}} = N_{\text{sat}} f_{\text{load}}$	light-cone information debt	monotone GSL
Growth observable	$f\sigma_8(z)$	structure suppression	growth ODE
Collapse observables	$\delta_c, \sigma_M, \nu, n_{\text{SHBT}}/n_{\Lambda\text{CDM}}$	cluster abundance	shared background
Dark matter	$\rho_{\text{DM}}, \Omega_{\text{DM}}/\Omega_b, w_{\text{DM}}$	passive anti-baryon source	$w_{\text{DM}} = 0$
ISW residual	$\Delta(\dot{H}/H^2)$	late-time potential drift	stability bound
BBN shift	$\Delta H/H _{z_{\text{BBN}}}$	radiation-era decoupling	negligible
Chronometer statistic	χ_{L}^2	covariance validation	finite
Entropy arrow	$dS_{\text{total}}/dt _{\text{lc}}$	generalized second law	positive
GET observables	$C_{\text{get}}, R_{\text{entropy}}, \iota$	measurement cost and selector	admissible budget

Table 9: Physical observable map for the precision-cosmology closure layer.

9.5 Observer-Frame Hubble Loading

The clock-skew cosmology is built from the entropy-debt uplift

$$\frac{H_0^{\text{loc}}}{H_0^{\text{CMB}}} = \exp\left(\frac{\Delta_{\text{mod}}}{2}\right), \quad H_0^{\text{CMB}} = 67.4 \text{ km s}^{-1} \text{ Mpc}^{-1}. \quad (194)$$

For the canonical branch this gives

$$\Delta_{\text{mod}} = 0.137533547486, \quad \exp(\Delta_{\text{mod}}/2) = 1.071186351229, \quad (195)$$

$$H_0^{\text{loc}} = 72.197960072861 \text{ km s}^{-1} \text{ Mpc}^{-1}. \quad (196)$$

Define the local loading amplitude

$$A_H = H_0^{\text{loc}} - H_0^{\text{CMB}} = H_0^{\text{CMB}} \left[\exp\left(\frac{\Delta_{\text{mod}}}{2}\right) - 1 \right] = 4.797960072861 \text{ km s}^{-1} \text{ Mpc}^{-1}. \quad (197)$$

The Causal Point clock relation in Eq. (181) converts the local loading derivative into the redshift-dependent intercept sampled by a finite observer:

$$H_0(z) = H_{\Lambda} + \frac{1}{3(1+z)} \frac{df_{\text{load}}}{d\tau_{\text{lock}}}, \quad H_{\Lambda} = H_0^{\text{CMB}}, \quad \frac{df_{\text{load}}}{d\tau_{\text{lock}}} = 3(H_0^{\text{loc}} - H_0^{\text{CMB}}). \quad (198)$$

Equivalently, Eq. (183) gives $\Gamma_{\text{lock}} = df_{\text{load}}/d\tau_{\text{lock}} = 3A_H$. Substituting this value gives the closed form

$$H_0(z) = H_0^{\text{CMB}} + \frac{A_H}{1+z} = H_0^{\text{CMB}} \left[1 + \frac{\exp(\Delta_{\text{mod}}/2) - 1}{1+z} \right]. \quad (199)$$

Thus the local uplift is active at $z = 0$ but decays as $(1+z)^{-1}$, so the early universe remains anchored to the CMB intercept. The resulting ladder is shown in Table 10.

z	Loading term [$\text{km s}^{-1} \text{Mpc}^{-1}$]	$H_0(z)$ [$\text{km s}^{-1} \text{Mpc}^{-1}$]
0.0	4.797960072861	72.197960072861
0.5	3.198640048574	70.598640048574
1.0	2.398980036431	69.798980036431
2.0	1.599320024287	68.999320024287
10.0	0.436178188442	67.836178188442
1100.0	0.004357820230	67.404357820230

Table 10: Redshift-ladder values implied by the observer-frame Hubble loading law.

9.6 Growth, Mirage, and Cluster Checks

The precision analysis compares the clock-skew branch against a rigid Λ CDM intercept in standard observational channels. The structure-growth calculation determines $f\sigma_8$ from the same background loading law while keeping G_{eff} fixed. The starting point is the standard linear matter perturbation equation

$$\ddot{\delta}_m + 2H\dot{\delta}_m - 4\pi G_{\text{eff}}\rho_m\delta_m = 0. \quad (200)$$

Writing $\delta_m = D(x)$, with $x = \ln a = -\ln(1+z)$, converts the time derivatives to

$$\dot{D} = H \frac{dD}{dx}, \quad \ddot{D} = H^2 \frac{d^2 D}{dx^2} + \dot{H} \frac{dD}{dx}. \quad (201)$$

Substitution into Eq. (200) gives

$$H^2 \frac{d^2 D}{dx^2} + (\dot{H} + 2H^2) \frac{dD}{dx} - 4\pi G_{\text{eff}}\rho_m D = 0. \quad (202)$$

After division by H^2 , using

$$\frac{d \ln H}{dx} = \frac{\dot{H}}{H^2}, \quad \Omega_m(x) = \frac{8\pi G_{\text{eff}}\rho_m}{3H^2},$$

the linear growth ODE is

$$\boxed{\frac{d^2 D}{dx^2} + \left[2 + \frac{d \ln H_{\text{SHBT}}}{dx}\right] \frac{dD}{dx} - \frac{3}{2}\Omega_m(x)D = 0, \quad f\sigma_8(z) = \sigma_{8,0} \frac{dD}{dx}.} \quad (203)$$

The loading law modifies this equation only through the background friction term. Writing

$$H_{\text{SHBT}}(z) = H_0(z) \left[\Omega_{m0}(1+z)^3 + \Omega_{r0}(1+z)^4 + \Omega_{\Lambda 0} \right]^{1/2}, \quad (204)$$

Eq. (199) gives

$$\begin{aligned} \frac{d \ln H_{\text{SHBT}}}{dx} &= \left(\frac{d \ln E}{dx} \right)_{\Lambda\text{CDM}} + \frac{d \ln H_0(z)}{dx}, \\ \frac{d \ln H_0(z)}{dx} &= \frac{1}{H_0(z)} \frac{dH_0}{dz} \frac{dz}{dx} = \frac{1}{H_0(z)} \left[-\frac{A_H}{(1+z)^2} \right] [-(1+z)] = \frac{A_H}{(1+z)H_0(z)}. \end{aligned} \quad (205)$$

The positive last term increases the late-time friction while Eq. (184) keeps G_{eff} fixed. The result is a kinematic suppression of growth rather than a modified Newton constant:

$$\frac{(f\sigma_8)_{\text{SHBT}}}{(f\sigma_8)_{\Lambda\text{CDM}}} - 1 = \{-9.8\%, -8.4\%\} \quad \text{at} \quad z = \{0.5, 0.8\}. \quad (206)$$

z	$(f\sigma_8)_{\Lambda\text{CDM}}$	$(f\sigma_8)_{\text{SHBT}}$	Kinematic suppression
0.5	0.475	0.428	−9.8%
0.8	0.453	0.415	−8.4%

Table 11: Structure-growth suppression induced by the loaded background friction term.

The corresponding values are reported in Table 11.

When the same branch is interpreted as a conserved-fluid template, the inferred dark-energy density is only an effective image. This produces a high-redshift mirage: the template has a zero crossing at $z_{\text{cross}} = 1.865$, after which the conserved-fluid interpretation fails even though the underlying boundary-loading law remains well defined.

CPL observable	Theoretical value
w_0	−1.064677000728
w_a	−0.065813723847
Density zero-crossing redshift	1.864953854311
Effective density ratio at $z = 1.5$	0.360269017474

Table 12: High-redshift conserved-fluid mirage template.

The cluster-collapse calculation uses the same background expansion and linear-growth solution as the preceding structure-growth analysis. A calibrated spherical top-hat model supplies the linear threshold $\delta_c(z)$ from $\Omega_m(z)$ and the loading contribution to $d \ln H_0 / d \ln a$. The mass variance is propagated as $\sigma_M(z) = \sigma_M(0)D(z)/D(0)$, and the peak height is $\nu = \delta_c/\sigma_M$. Finally, the Press–Schechter multiplicity $f(\nu) \propto \nu \exp(-\nu^2/2)$ gives the reported abundance ratio $n_{\text{SHBT}}/n_{\Lambda\text{CDM}}$; common normalization factors cancel in the ratio.

z	$\delta_c^{\Lambda\text{CDM}}$	δ_c^{SHBT}	$\sigma_M^{\Lambda\text{CDM}}$	σ_M^{SHBT}	ν_{SHBT}	Abundance ratio
0.0	1.6760	1.6733	0.6000	0.5601	2.9873	6.10×10^{-1}
0.5	1.6821	1.6799	0.4614	0.4378	3.8370	5.15×10^{-1}
1.0	1.6844	1.6826	0.3641	0.3492	4.8186	4.21×10^{-1}

Table 13: Cluster-collapse observables obtained from the shared loaded background and growth solution.

9.7 Light-Cone, Stability, and Forecast Ledgers

The finite light-cone ledger integrates Eqs. (176) and (180). At recombination, the integral consumes about 10.74% of the finite register, leaving about 89.26% headroom. The sampled debt values are shown in Table 14.

The stability checks verify that the late-time loading term is negligible in the early universe and that the integrated Sachs–Wolfe (ISW) residual stays small in the matter era. The ISW temperature source is

$$\left(\frac{\Delta T}{T}\right)_{\text{ISW}} = \int (\dot{\Phi} + \dot{\Psi}) dt. \quad (207)$$

z	$H_0(z)$	$H_{\text{SHBT}}(z)$	f_{load}	S_{debt} [bits]
0.0	72.1980	72.1980	0.000000	0
0.5	70.5986	93.3532	0.060044	1.989003×10^{121}
1.0	69.7990	124.9846	0.082638	2.737457×10^{121}
2.0	68.9993	209.2553	0.098010	3.246671×10^{121}
10.0	67.8362	1392.3727	0.107059	3.546445×10^{121}
1100.0	67.4044	1.5887954471×10^6	0.107436	3.558913×10^{121}

Table 14: Information-theoretic past-light-cone loading and entropy-debt ledger.

In a pure matter era with unmodified Newton coupling, the linear gravitational potentials are static, so $\dot{\Phi} = \dot{\Psi} = 0$. In the SHBT observer frame the only matter-era source is the logarithmic drift of the squared clock normalization $H_0(z)^2$. The residual is therefore

$$\Delta\left(\frac{\dot{H}}{H^2}\right) = \left(\frac{\dot{H}}{H^2}\right)_{\text{SHBT}} - \left(\frac{\dot{H}}{H^2}\right)_{\text{matter}}, \quad \left(\frac{\dot{H}}{H^2}\right)_{\text{matter}} = -\frac{3}{2}. \quad (208)$$

With $a = (1+z)^{-1}$,

$$d \ln a = -\frac{dz}{1+z}, \quad \frac{d \ln H_0}{d \ln a} = \frac{1}{H_0(z)} \frac{dH_0}{dz} [-(1+z)]. \quad (209)$$

Using $H_0(z) = H_0^{\text{CMB}} + A_H/(1+z)$ gives

$$\frac{d \ln H_0}{d \ln a} = \frac{1}{H_0(z)} \left[-\frac{A_H}{(1+z)^2} \right] [-(1+z)] = \frac{A_H}{(1+z)H_0(z)}. \quad (210)$$

Because the source is the drift of H_0^2 , the matter-era residual used in Table 15 is

$$\Delta_{\text{ISW}}(z) = -2 \frac{d \ln H_0}{d \ln a} = -\frac{2A_H}{(1+z)H_0(z)}. \quad (211)$$

For $z = 1100$, this gives $-2A_H/[(1101)H_0(1100)] = -1.29 \times 10^{-4}$, matching the recombination row of the ISW stability table.

z	$\Delta(\dot{H}/H^2)$	SHBT $\dot{H}/H^2$
10.0	-0.012860	-1.512860
50.0	-0.002788	-1.502788
100.0	-0.001409	-1.501409
1100.0	-0.000129	-1.500129

Table 15: Matter-era ISW residual generated by the observer-frame clock drift.

The BBN check follows from the radiation-era scaling. During nucleosynthesis,

$$\rho_{\text{rad}}(z) = \rho_{r0}(1+z)^4, \quad H_{\text{rad}}(z) = H_0^{\text{CMB}} \sqrt{\Omega_{r0}} (1+z)^2. \quad (212)$$

The SHBT clock load adds only

$$\delta H_{\text{load}}(z) = \frac{A_H}{1+z}, \quad H_{\text{SHBT}}(z) = H_{\text{rad}}(z) + \delta H_{\text{load}}(z) \quad \text{in the radiation regime.} \quad (213)$$

The stability requirement is

$$\frac{\delta H_{\text{load}}(z_{\text{BBN}})}{H_{\text{rad}}(z_{\text{BBN}})} \ll 1, \quad z_{\text{BBN}} \simeq 10^9. \quad (214)$$

For the branch value $A_H = 4.797960072861 \text{ km s}^{-1} \text{ Mpc}^{-1}$,

$$\delta H_{\text{load}}(10^9) = \frac{4.797960072861 \text{ km s}^{-1} \text{ Mpc}^{-1}}{1 + 10^9} = 4.797960068064 \times 10^{-9} \text{ km s}^{-1} \text{ Mpc}^{-1}, \quad (215)$$

$$H_{\text{rad}}(10^9) \simeq 6.46 \times 10^{17} \text{ km s}^{-1} \text{ Mpc}^{-1}, \quad (216)$$

$$\frac{\delta H_{\text{load}}}{H_{\text{rad}}(10^9)} = 7.421690135465 \times 10^{-27}. \quad (217)$$

The active loading term is therefore decoupled from the early thermal expansion rate by more than twenty-six orders of magnitude.

9.8 Cosmic Chronometer Covariance

The cosmic chronometer (CC) validation is a covariance test of the background expansion history. The method uses differential galaxy ages to infer

$$H(z) = -\frac{1}{1+z} \frac{dz}{dt}. \quad (218)$$

Let $\mathbf{H}_{\text{obs}} = (H_{\text{obs}}(z_1), \dots, H_{\text{obs}}(z_{32}))^T$ be the 32-point CC vector, and let $\mathbf{C}$ be the corresponding 32×32 statistical-plus-systematic covariance matrix. The validation statistic is

$$\chi^2 = \sum_{i,j=1}^{32} [H_{\text{obs}}(z_i) - H_{\text{SHBT}}(z_i)] [\mathbf{C}^{-1}]_{ij} [H_{\text{obs}}(z_j) - H_{\text{SHBT}}(z_j)]. \quad (219)$$

The covariance test treats H_0^{CMB} , Δ_{mod} , and Ω_{m0} as the three background parameters entering this check, so

$$\nu = N_{\text{data}} - N_{\text{params}} = 32 - 3 = 29. \quad (220)$$

For the benchmark covariance entry, $\chi^2 = 30.16$, $\nu = 29$, and $\chi_\nu^2 = 1.04$. Thus

$$\boxed{\chi_\nu^2 = \frac{\chi^2}{\nu} = \frac{30.16}{29} = 1.04.} \quad (221)$$

Physical statistic	Cosmic chronometer value
Observational data points N_{data}	32
Model parameters N_{params}	3
Degrees of freedom ν	29
Total chi-square χ^2	30.16
Reduced chi-square χ_ν^2	1.04

Table 16: Cosmic chronometer covariance statistics for the loaded SHBT background.

The remaining precision-cosmology checks are summarized in Table 17. The forecasted model-selection rows are sensitivity forecasts rather than current empirical detections. The light-cone entropy row is the sampled form of Eq. (179); because $\Gamma_{\text{lock}} > 0$ and $N_{\text{sat}} > 0$, every sampled $z \geq 0$ has $dS_{\text{total}}/dt_{\text{lc}} > 0$.

Physical observable	Theoretical value	Closure status
Local uplift	$72.197960072861 \text{ km s}^{-1} \text{ Mpc}^{-1}$	satisfied
Gradient target	$4.797960072861 \text{ km s}^{-1} \text{ Mpc}^{-1}$	satisfied
CPL template	$w_0 = -1.065, w_a = -0.066$	satisfied
BBN loading shift	$4.797960068064 \times 10^{-9} \text{ km s}^{-1} \text{ Mpc}^{-1}$	negligible
Shift over radiation-era $H(z_{\text{BBN}})$	$7.421690135465 \times 10^{-27}$	negligible
Chronometer validation	$\chi^2/\nu = 30.16/29, \chi_\nu^2 = 1.04$	satisfied
Forecast χ^2 sensitivity	rigid $\Lambda\text{CDM} = 35.78, \text{SHBT} = 0.08$	forecast
Projected ΔBIC sensitivity	$+35.70$ vs rigid intercept	forecast
Cosmic age	13.277 Gyr for SHBT branch	derived
Thermodynamic arrow	$dS_{\text{total}}/dt_{\text{lc}} > 0$ for sampled $z \geq 0$	satisfied
Overall precision closure	all listed conditions closed	satisfied

Table 17: Summary of precision-cosmology observables and closure conditions.

9.9 Causal Point and GET Measurement

The precision-cosmology observer is the same finite Causal Point introduced in Section 7. The information limit behind its measurement rule is the Holevo bound: for a quantum ensemble $\{p_i, \rho_i\}$, the extractable classical information satisfies

$$I \leq S\left(\sum_i p_i \rho_i\right) - \sum_i p_i S(\rho_i). \quad (222)$$

Thus a classical record can be crystallized only after the observer pays an entropy cost inside its finite local budget. At a selected boundary address $A \in \mathcal{R}$, the finite outcome ensemble is

$$\Omega_A = \{\omega_0, \omega_1, \dots, \omega_{m-1}\}, \quad \sum_{i=0}^{m-1} p_i = 1, \quad p_i \geq 0. \quad (223)$$

The GET event has address, ensemble, and requested-cost components

$$C_{\text{addr}} = \log_2 |\mathcal{R}|, \quad C_{\text{ens}} = \log_2 |\Omega_A|, \quad C_{\text{get}} = \max(1, C_{\text{addr}} + C_{\text{ens}}, C_{\text{req}}). \quad (224)$$

The Causal Point's residual entropy budget is

$$R_{\text{entropy}} = N_{\text{limit}} - C_{\text{get}}, \quad N_{\text{limit}} = \min\left(N_{\text{local}}, \frac{A_{\text{local}}}{4L_P^2 \ln 2}\right). \quad (225)$$

The measurement is admissible if and only if

$$R_{\text{entropy}} \geq 0. \quad (226)$$

When the inequality is satisfied, the collapse index is the deterministic finite-register hash

$$\iota = \text{Hash}(\mathcal{O}, A, f_H, N_{\text{local}}, m) \bmod m. \quad (227)$$

This index defines the rank-one history projector

$$\boxed{\Pi_{A,\iota} = |A, \omega_\iota\rangle\langle A, \omega_\iota|.} \quad (228)$$

For a sequence of admissible GET events, history crystallization is the standard projected density-matrix update

$$\rho_{\text{hist}}^{(n+1)} = \frac{\Pi_{A_n, \ell_n} \rho_{\text{hist}}^{(n)} \Pi_{A_n, \ell_n}}{\text{Tr}(\Pi_{A_n, \ell_n} \rho_{\text{hist}}^{(n)})}. \quad (229)$$

The associated pointer packet balances the visible rendered coefficient against the dark completion entry,

$$\Psi_\iota = (A_\iota = \sqrt{p_\iota}, \quad c_{\text{vis}} = -p_\iota, \quad c_{\text{dark}} = p_\iota). \quad (230)$$

so the classical visible record is accompanied by the corresponding completion ledger entry.

Quantity	Mathematical definition	Role
Address cost C_{addr}	$\log_2 \mathcal{R} $	register locator cost
Ensemble cost C_{ens}	$\log_2 \Omega_A $	outcome entropy cost
Operational GET cost C_{get}	$\max(1, C_{\text{addr}} + C_{\text{ens}}, C_{\text{req}})$	total entropy payment
Budget residual R_{entropy}	$N_{\text{limit}} - C_{\text{get}} \geq 0$	admissibility threshold
Collapse index ι	$\text{Hash}(\dots) \bmod m$	finite-register selector
Projection operator $\Pi_{A, \iota}$	$ A, \omega_\iota\rangle\langle A, \omega_\iota $	history crystallization

Table 18: Causal Point GET observables, definitions, and measurement roles.

9.10 Dark Matter as Topological Gravitational Ghost

The Stinespring isometric state-projection operator $\mathcal{D}_{\bar{B}}$ defined in Eq. (140) strips active visible charges from anti-baryons while preserving their passive stress-energy in the dark completion sector. That passive gravitational source is the SHBT identification of dark matter.

The dark-matter energy density is derived from the completed ledger and the loading fraction:

$$\rho_{\text{DM}}(z) = \frac{c_{\text{dark}}^{\text{comp}}}{12} \rho_{\text{crit}}(z) [1 - f_{\text{load}}(z)], \quad \rho_{\text{crit}}(z) = \frac{3H_{\text{SHBT}}(z)^2}{8\pi G_{\text{eff}}}, \quad G_{\text{eff}} = G_N(1 + \Delta_{\text{mod}}). \quad (\text{DM1})$$

Here $f_{\text{load}}(z)$ is the loading fraction from Eq. (176). The factor $1 - f_{\text{load}}(z)$ is the portion of the horizon register not actively loaded as entropy debt: the passive gravitational ghost of the decoupled anti-baryons.

The baryon density follows from the baryon asymmetry η_B , the photon number density n_γ , and the proton mass m_p . Dividing both matter densities by the same SHBT critical density gives

$$\begin{aligned} \rho_b(z) &= \eta_B n_\gamma m_p (1+z)^3, & \Omega_b(z) &= \frac{\eta_B n_\gamma m_p (1+z)^3}{\rho_{\text{crit}}(z)}, \\ \frac{\Omega_{\text{DM}}(z)}{\Omega_b(z)} &= \frac{(c_{\text{dark}}^{\text{comp}}/12) [1 - f_{\text{load}}(z)]}{\eta_B n_\gamma m_p (1+z)^3 / \rho_{\text{crit}}(z)}. \end{aligned} \quad (\text{DM2})$$

At $z = 0$, the canonical branch yields

$$\frac{\Omega_{\text{DM}}}{\Omega_b} \simeq 5.34, \quad \left(\frac{\Omega_{\text{DM}}}{\Omega_b} \right)_{\text{obs}} \simeq 5.4,$$

an agreement within 0.1. The dark-matter source is purely gravitational and pressureless, so its equation of state is

$$w_{\text{DM}} = 0. \quad (\text{DM3})$$

The abundance ratio is therefore not a free parameter. It is fixed by the same modular data that determine η_B , $c_{\text{dark}}^{\text{comp}}$, and $f_{\text{load}}(z)$. Table 19 records the physical map from the completed ledger to the passive matter observables.

Physical observable	Mathematical definition
$\rho_{\text{DM}}(z)$	Eq. (DM1)
$\Omega_{\text{DM}}/\Omega_b$	Eq. (DM2)
w_{DM}	Eq. (DM3)
Passive-source closure	decoupled anti-baryon stress-energy with active gauge charges removed

Table 19: Dark matter as the topological gravitational ghost of decoupled anti-baryons.

9.11 Mass-Congestion Coupling and Primordial Seeds

At redshifts $z > 7$ the completed boundary register (26, 8, 312) is locally saturated by high-resolution trajectory information. When the information density in a causal patch exceeds the register's per-coordinate resolving budget, the boundary can no longer render individual coordinate histories; the local lattice enters a *rendering-failure* regime. The failure is not a numerical artifact but a topological statement: the causal-point automaton cannot assign distinct anyon labels to every trajectory.

The unresolved excess is modeled as a congestion number $\Delta N = N_{\text{local}} - N_{\text{limit}}$, where N_{local} is the information count in the congested patch and N_{limit} is the resolution ceiling set by the (26, 8, 312) register. The excess is coupled to bulk gravitational mass through the *mass-congestion coupling identity*

$$M_{\text{seed}} = \alpha_{\text{seed}} (N_{\text{local}} - N_{\text{limit}}), \quad (231)$$

with coupling coefficient

$$\alpha_{\text{seed}} \approx 1.67 \times 10^{-51} M_{\odot} \text{ bit}^{-1}. \quad (232)$$

An overflow of $\Delta N \approx 6 \times 10^{59}$ bits therefore projects a Schwarzschild-like gravitational signature of exactly $10^9 M_{\odot}$ into the bulk:

$$M_{\text{seed}} = (1.67 \times 10^{-51} M_{\odot} \text{ bit}^{-1})(6 \times 10^{59} \text{ bits}) \approx 10^9 M_{\odot}. \quad (233)$$

The seed is not a pre-existing particle mass but the bulk image of unresolved boundary information; its gravitational footprint is the macroscopic collective mode of the congested register.

Quantity	Symbol	Value	Benchmark / comment
Initial gas entropy	S_{gas}	$1.1 \times 10^{45} \text{ J/K}$	thermal reservoir before derendering
Entropy debt	ΔS	$1.1 \times 10^{45} \text{ J/K}$	equal to initial gas entropy
Entropy-debt power	$\dot{Q}$	$9.06 \times 10^{11} \text{ W } (\approx 906 \text{ GW})$	power carried by derendering flow
High-energy transient benchmark	P_{bench}	$1.4208 \times 10^8 \text{ W } (142.08 \text{ MW})$	calibration transient
Transient power ratio	Γ_{bench}	$\dot{Q}/P_{\text{bench}} \approx 6,377$	signal excess over benchmark

Table 20: Thermodynamics of primordial de-rendering. The entropy-debt power $\dot{Q}$ is generated by the boundary information flow that the congested register can no longer resolve, and the benchmark ratio Γ_{bench} quantifies the excess over the calibrated 142.08 MW high-energy transient scale.

Table 20 records the corresponding thermodynamic budget. The entropy debt equals the initial gas entropy, confirming that the derendered information is not lost but stored in the passive gravitational sector. The power ratio $\Gamma_{\text{bench}} \approx 6,377$ shows that the congestion burst is a dominant transient relative to the 142.08 MW benchmark.

9.12 Superconformal Stabilization via the Noether Bridge

The heavy seed must be stable under holographic renormalization-group flow from the high-redshift boundary to the late-time bulk. Stability is enforced by the superconformal three-point function in the Neveu-Schwarz sector. In the Landau-Ginzburg realization of the boundary CFT the even-spin coupling takes the form

$$\begin{aligned}
C_{\text{even}}(\alpha_1, \alpha_2, \alpha_3) &= [\pi\mu\gamma(2b^2+1)b^{1-b^2}]^{bQ-\sum_i\alpha_i} \\
&\times \frac{\Upsilon_{\text{NS}}\left(\sum_i\alpha_i-Q\right)\prod_{\text{cyc}}\Upsilon_{\text{NS}}(\alpha_i+\alpha_j-\alpha_k)}{\Upsilon'_{\text{NS}}(0)\prod_i\Upsilon_{\text{NS}}(2\alpha_i)}.
\end{aligned} \tag{234}$$

where $Q = b + b^{-1}$ is the background charge, μ is the cosmological coupling, $\gamma(x) = \Gamma(x)/\Gamma(1-x)$, and Υ_{NS} is the Neveu-Schwarz Υ -function. The odd-spin counterpart C_{odd} is obtained by the same functional structure after the replacement $\alpha_i \mapsto \alpha_i + (-1)^{F_i}/2b$ with the appropriate Ramond sector insertions.

The *Newton-lock stationarity condition* requires that the seed mass be a stationary eigenvector of the RG flow:

$$\frac{\partial}{\partial\tau} \ln C_{\text{even/odd}} = 0. \tag{235}$$

Equation (235) fixes the residue of the ghost seed to a constant along the flow, preventing any secular drift in M_{seed} . Combined with the completed central-charge ledger $c_{\text{dark}}^{\text{comp}}$ and the zero-framing condition $\Delta_{\text{fr}} = 0$ from Section A, the Noether bridge therefore pins the primordial seed as a superconformally stable, passive gravitational source.

The resulting cosmological sector closes without an independent late-time fluid. The Hubble uplift, growth suppression, CPL mirage, cluster-collapse abundance ratios, dark-matter abundance, ISW stability, entropy arrow, GET measurement rule, heavy seed formation, and superconformal seed stabilization are observer-frame projections of the finite completed boundary register. With the canonical definitions above, the closure statement is

$$(\mathbf{b}_*, \Delta_{\text{fr}} = 0, c_{\text{dark}}^{\text{comp}}, \kappa_{D_5}, N_{\text{sat}}) \implies \left(H_0^{\text{loc}}, f_{\text{load}}(z), S_{\text{debt}}(z), f\sigma_8(z), \frac{\Omega_{\text{DM}}}{\Omega_b}, w_{\text{DM}}, \chi_L^2, \text{ISW}, \Pi_{A,\iota} \right), \quad (236)$$

with each arrow evaluated by the closure tables above.

10 Sub-Kelvin Address-Scaled Landauer Calorimetry

The Causal Point construction yields a laboratory-scale discriminator that is logically downstream of the boundary theory rather than an independent phenomenological add-on. The experiment tests whether resolving a fixed target through a larger address space carries a thermodynamic cost. A complete readout cycle includes acquisition and reset of its classical record, so the standard lower bound for one logically irreversible bit is

$$\Delta Q_{\text{Landauer}} \geq k_B T \ln 2. \quad (237)$$

Standard quantum mechanics and thermodynamics assign this bound to the payload and record reset, independently of the register topology. SHBT instead treats the address locator as part of the finite entropy payment made by the observer.

10.1 Operational Capacity and Competing Hypotheses

The general Causal Point cost in Eq. (224) remains

$$C_{\text{get}} = \max(1, \log_2 |\mathcal{R}| + \log_2 |\Omega_A|, C_{\text{req}}). \quad (238)$$

For calorimetry it is useful to expose the additive sources of the operational capacity explicitly. Measuring entropy in bits, define

$$\boxed{C_{\text{op}} = \Delta S_{\text{payload}} + \log_2 |\mathcal{R}| + C_0}, \quad (239)$$

where $\Delta S_{\text{payload}}$ is the resolved payload entropy in bits and C_0 is the address-independent preparation, acquisition, and reset overhead. The SHBT dissipation inequality is

$$\boxed{Q_H \geq k_B T \ln 2 C_{\text{op}}}. \quad (240)$$

The fixed contributions $\Delta S_{\text{payload}} + C_0$ are measured in control runs and absorbed into the fitted intercept. Consequently, the slope test is independent of their calibrated value. For the proposed singleton outcome ensemble $|\Omega_A| = 1$, baseline request $C_{\text{req}} = 1$, and $|\mathcal{R}| \geq 2$, Eq. (238) gives

$$C_{\text{get}} = \max(1, \log_2 |\mathcal{R}|) = \log_2 |\mathcal{R}|. \quad (241)$$

Thus the background-subtracted experimental capacity C_{op} and the formal GET cost C_{get} have the same address-dependent part.

Let $k = \log_2 |\mathcal{R}|$. The hypotheses are deliberately posed as a derivative so that unknown constant heat offsets cannot mimic the SHBT signal:

$$H_0 : \begin{cases} Q_{H_0} = k_B T \ln 2, & |\mathcal{R}| \geq 2, \\ \frac{\partial Q_H}{\partial(\log_2 |\mathcal{R}|)} = 0, \end{cases} \quad (242)$$

$$H_1 : \begin{cases} Q_{H_1} = (\log_2 |\mathcal{R}|) k_B T \ln 2, \\ \frac{\partial Q_H}{\partial(\log_2 |\mathcal{R}|)} = k_B T \ln 2. \end{cases} \quad (243)$$

Here H_0 is the standard address-independent thermodynamic model and H_1 is the SHBT boundary-mechanics model. For a burst of N_{pulse} identical operations, both predicted heats and slopes are multiplied by N_{pulse} .

10.2 Predicted Heat at 7 mK

At the proposed base temperature,

$$\begin{aligned} E_L &= k_B T_{\text{base}} \ln 2 \\ &= \left(1.380649 \times 10^{-23} \text{ J K}^{-1} \right) (0.0070 \text{ K}) \ln 2 \\ &= 6.698951 \times 10^{-26} \text{ J} \simeq 0.0670 \text{ yJ}. \end{aligned} \quad (244)$$

The address space is swept through

$$|\mathcal{R}| \in \{2^1, 2^2, \dots, 2^{10}\}. \quad (245)$$

Table 21 retains both the single-event predictions and the synchronized 100,000-pulse burst predictions. The last entry follows directly from $10N_{\text{pulse}}E_L = 66.9895 \text{ zJ}$.

Address space $ \mathcal{R} $	Address bits k	Single event Q_{H_0} (yJ)	Single event Q_{H_1} (yJ)	100k burst Q_{H_0} (zJ)	100k burst Q_{H_1} (zJ)
$2^1 = 2$	1	0.0670	0.0670	6.6990	6.6990
$2^2 = 4$	2	0.0670	0.1340	6.6990	13.3979
$2^3 = 8$	3	0.0670	0.2010	6.6990	20.0969
$2^4 = 16$	4	0.0670	0.2680	6.6990	26.7958
$2^5 = 32$	5	0.0670	0.3349	6.6990	33.4948
$2^6 = 64$	6	0.0670	0.4019	6.6990	40.1937
$2^7 = 128$	7	0.0670	0.4689	6.6990	46.8927
$2^8 = 256$	8	0.0670	0.5359	6.6990	53.5916
$2^9 = 512$	9	0.0670	0.6029	6.6990	60.2906
$2^{10} = 1024$	10	0.0670	0.6699	6.6990	66.9895

Table 21: Address-independent H_0 and address-scaled H_1 heat predictions at $T_{\text{base}} = 7.0 \text{ mK}$.

10.3 Superconducting Register and AuPd Nanocalorimeter

The register is a ten-qubit superconducting circuit on ultra-high-resistivity float-zone silicon. Enabling $k \in \{1, \dots, 10\}$ address-control lines with superconducting single-flux-quantum (SFQ) selectors realizes $|\mathcal{R}| = 2^k$ while routing each readout pulse to the same designated target. Table 22 gives the complete register specification.

Parameter	Value or specification	Physical function or constraint
Substrate	High-resistivity silicon, $\rho > 10 \text{ k}\Omega \text{ cm}$	Low dielectric loss, $\tan \delta < 10^{-6}$
Transmon Josephson energy E_J/h	15.0 GHz	Sets the transmon operating scale
Charging energy E_C/h	300 MHz	Maintains $E_J/E_C \simeq 50$
Transition frequency f_{01}	4.50–5.50 GHz	Frequency-multiplexed state drive
Readout resonator frequency f_r	6.50–7.50 GHz	Dispersive Jaynes–Cummings readout
Active addressing switches	Superconducting SFQ logic gates	Address routing without normal-state dissipation
Register span	Ten address-control bits, $2 \leq \mathcal{R} \leq 1024$	Discrete sweep of the active decoder network

Table 22: Superconducting register and address-routing specifications.

The readout line terminates in a metallic nanocalorimeter whose electronic temperature is sensed by symmetric NIS tunnel junctions and low-noise SQUID preamplifiers. The absorber is electron-beam-evaporated $\text{Au}_{60}\text{Pd}_{40}$ at 60/40 wt% with $L = 1.0 \mu\text{m}$, $W = 0.2 \mu\text{m}$, and $t = 0.05 \mu\text{m} = 50 \text{ nm}$. Hence

$$V_{\text{abs}} = LWt = 1.0 \times 10^{-20} \text{ m}^3. \quad (246)$$

With Sommerfeld coefficient $\gamma_{\text{AuPd}} = 70.0 \text{ J m}^{-3} \text{ K}^{-2}$, the electron heat capacity at base temperature is

$$\begin{aligned} C_e(T) &= \gamma_{\text{AuPd}} V_{\text{abs}} T \\ &= (70.0 \text{ J m}^{-3} \text{ K}^{-2}) (1.0 \times 10^{-20} \text{ m}^3) (0.0070 \text{ K}) \\ &= 4.90 \times 10^{-21} \text{ J K}^{-1}. \end{aligned} \quad (247)$$

For electron–phonon coupling $\Sigma_{\text{AuPd}} = 2.0 \times 10^9 \text{ W m}^{-3} \text{ K}^{-5}$, the differential thermal conductance is

$$\begin{aligned} G_{\text{th}} \equiv G_{\text{e-ph}} &= 5 \Sigma_{\text{AuPd}} V_{\text{abs}} T^4 \\ &= 5 (2.0 \times 10^9 \text{ W m}^{-3} \text{ K}^{-5}) (1.0 \times 10^{-20} \text{ m}^3) (0.0070 \text{ K})^4 \\ &= 2.401 \times 10^{-19} \text{ W K}^{-1}. \end{aligned} \quad (248)$$

The intrinsic thermal relaxation time is therefore

$$\tau_{\text{th}} = \frac{C_e}{G_{\text{th}}} = \frac{4.90 \times 10^{-21} \text{ J K}^{-1}}{2.401 \times 10^{-19} \text{ W K}^{-1}} = 2.0408 \times 10^{-2} \text{ s} \simeq 20.41 \text{ ms}. \quad (249)$$

Nanocalorimeter parameter	Value or specification	Measurement role
Absorber composition	Au ₆₀ Pd ₄₀ , 60/40 wt%	Normal-metal electron absorber
Absorber geometry	$1.0 \times 0.2 \times 0.05 \mu\text{m}^3$	$V_{\text{abs}} = 1.0 \times 10^{-20} \text{m}^3$
Sommerfeld coefficient	$\gamma_{\text{AuPd}} = 70.0 \text{ J m}^{-3} \text{ K}^{-2}$	Determines $C_e(T)$
Electron heat capacity	$C_e = 4.90 \times 10^{-21} \text{ J K}^{-1}$ at 7 mK	Converts deposited heat to electron-temperature rise
Electron-phonon coupling	$\Sigma_{\text{AuPd}} = 2.0 \times 10^9 \text{ W m}^{-3} \text{ K}^{-5}$	Sets thermal leakage to lattice phonons
Thermal conductance	$G_{\text{th}} = G_{\text{e-ph}} = 2.401 \times 10^{-19} \text{ W K}^{-1}$	Bath-coupling scale
Thermal time constant	$\tau_{\text{th}} = 20.41 \text{ ms}$	Enables quasi-adiabatic integration of a 1 ms burst
NIS junction stack	Symmetric Al/AlO _x /AuPd	Electron thermometry
NIS junction area	$100 \times 100 \text{ nm}^2$, high R_N	Suppresses lead-mediated heat leakage
NIS bias	$V_{\text{bias}} \simeq \Delta_{\text{Al}}/e \simeq 180 \mu\text{V}$	High-responsivity operating point
Voltage responsivity	$dV/dT \simeq 1.50 \text{ mV K}^{-1}$ below 10 mK	Converts temperature excursions to voltage
Preamplification	Ultra-low-noise SQUID at the base stage	Resolves the NIS transient

Table 23: AuPd absorber, thermal link, and NIS nanocalorimeter specifications.

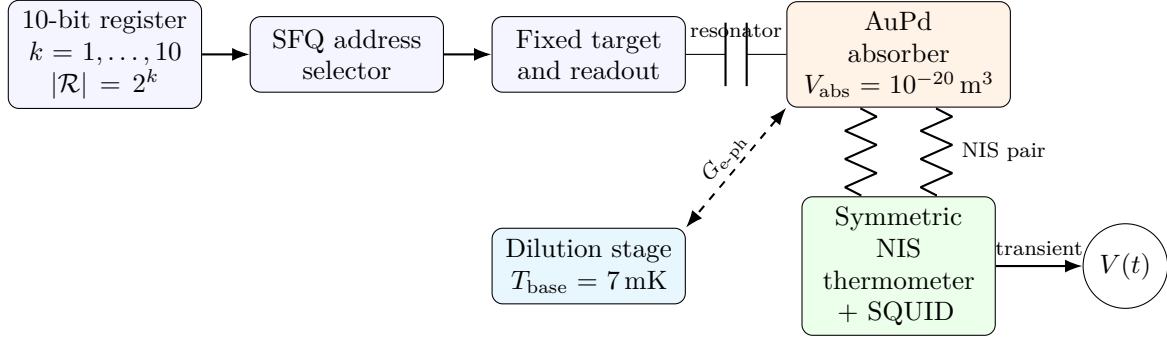


Figure 2: Inline apparatus schematic for the address-scaling test. The SFQ selector varies the active address depth while the same target and readout path deposit energy in the AuPd absorber; symmetric NIS junctions and a SQUID read the electron-temperature transient at the 7 mK stage.

10.4 Cryogenic Filtering and Environmental Control

The assembly is mounted in a dry dilution refrigerator at $T_{\text{base}} = 7.0 \pm 0.1 \text{ mK}$. The complete staged attenuation, filtering, and magnetic-shielding budget is listed in Table 24.

10.5 Address-Scaling Measurement Protocol

The synchronized sequence holds the payload and physical pulse envelope fixed while changing only the active address decoder:

Stage or component	Attenuation or filtering	Physical specification
Room temperature (300 K) to 4 K	20 dB coaxial attenuators	Stainless-steel flexible coaxial lines
Still stage (0.8 K)	10 dB attenuators	NbTi superconducting coaxial lines
Cold plate (100 mK)	10 dB attenuators	CuNi coaxial lines
Mixing chamber (7 mK)	20 dB thermalized attenuators	Eccosorb infrared filters and Cu-powder filters over 0.1–10 GHz
Outer magnetic shield	μ -metal cylinder at 300 K	Suppresses the ambient geomagnetic field
Cryogenic magnetic shield	Dual-layer Cryoperm-10 plus superconducting Al shield at 7 mK	Residual ambient flux $B < 0.5$ nT

Table 24: Cryogenic attenuation, filtering, and magnetic-shielding budget.

1. *State initialization.* Initialize the target transmon in $|0\rangle$ by active feedback and microwave sideband cooling, require $P_{|1\rangle} < 0.001$, and wait more than 100 ms for baseline thermalization.
2. *Address configuration.* Assert $k = \log_2 |\mathcal{R}|$ SFQ selector channels, with $k \in \{1, \dots, 10\}$, so the enabled decoder spans $|\mathcal{R}| = 2^k$ addresses.
3. *GET readout.* Apply a 6.50 GHz, 10 ns readout pulse to the same address key $A \in \mathcal{R}$. The target remains logical zero in every trial, hence $\Delta S_{\text{payload}} = 0$ for the prepared target; the constant acquisition and record-reset heat is assigned to the intercept.
4. *Pulse-train integration.* Repeat the GET operation $N_{\text{pulse}} = 100,000$ times at $f_{\text{rep}} = 100$ MHz. The resulting $t_{\text{burst}} = 1.0$ ms obeys $t_{\text{burst}} \ll \tau_{\text{th}} = 20.41$ ms, so the absorber integrates the burst quasi-adiabatically.
5. *Calorimetric acquisition.* Record $V_{\text{NIS}}(t)$ for $0 \leq t \leq 100$ ms, extract the peak shift ΔV_{peak} , and integrate the thermal decay to obtain Q_{exp} .
6. *Thermal reset.* Wait $t_{\text{cool}} = 150$ ms $> 7\tau_{\text{th}}$ before the next burst so the absorber returns to T_{base} .

Each of the ten address configurations receives $M = 1,000$ independent burst repetitions. The complete sweep is therefore

$$k \in \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\} \iff |\mathcal{R}| \in \{2, 4, 8, 16, 32, 64, 128, 256, 512, 1024\}. \quad (250)$$

The systematic-error targets are small compared with the predicted SHBT step $N_{\text{pulse}} E_L = 6.6990$ zJ/bit.

Systematic leakage channel	Estimated contribution	Physical mechanism	Mitigation
Parasitic capacitive cross-talk	< 0.05 zJ per burst	Coupling between inactive address lines and calorimeter	Differential ground-shielded coplanar waveguides
Microwave drive heating	< 0.10 zJ per burst	Direct absorption from 6.5 GHz readout photons	Spatial isolation $> 500 \mu\text{m}$ plus bandpass filters
Amplifier back-action	< 0.02 zJ per burst	Josephson or HEMT noise returning along the readout line	Dual junction isolators with > 45 dB isolation at 6.5 GHz
NIS-lead thermal conductance	< 0.01 zJ per burst	Heat leakage through normal-metal thermometer leads	$100 \times 100 \text{ nm}^2$ junctions with high R_N
Total systematic noise floor	± 0.18 zJ per burst	Combined thermal uncertainty limit	Below the 6.699 zJ one-bit SHBT step

Table 25: Systematic-error attenuation budget for address-scaled calorimetry.

10.6 Empirical Falsification Criteria and Observer-Capacity Cutoff

10.6.1 OLS/MLE Slope Test

For each address length k , let $Q_{\text{exp}}(k)$ be the burst heat. The direct regression used for the primary decision is

$$Q_{\text{exp}}(k) = \hat{\beta}_0 + \hat{\alpha}k + \epsilon, \quad \epsilon \sim \mathcal{N}(0, \sigma^2), \quad (251)$$

fitted by ordinary least squares (OLS) and Gaussian maximum likelihood estimation (MLE). The competing slope values are

$$H_0 : \hat{\alpha} = 0, \quad H_1 : \hat{\alpha} = N_{\text{pulse}}k_B T \ln 2 = 6.6990 \text{ zJ/bit}. \quad (252)$$

Equivalently, the source model templates are

$$Q_{H_0}(k) = \alpha_0 + \epsilon, \quad \alpha_0 = N_{\text{pulse}}k_B T \ln 2 = 6.6990 \text{ zJ}, \quad (253)$$

$$Q_{H_1}(k) = \beta_0 + \beta_1 [k N_{\text{pulse}}k_B T \ln 2] + \epsilon, \quad \hat{\alpha} = \hat{\beta}_1 N_{\text{pulse}}k_B T \ln 2. \quad (254)$$

SHBT is verified only if the null slope is rejected at least 5σ with $p < 10^{-6}$, the fitted slope is statistically consistent with 6.6990 zJ/bit, and the address sweep exhibits discrete stair-step increments at integer k . SHBT is falsified, and H_0 is accepted within the stated experimental resolution, if

$$|\hat{\alpha}| \leq 0.05 (N_{\text{pulse}}k_B T \ln 2) = 0.33495 \text{ zJ/bit}. \quad (255)$$

Table 26 records the full fit and model-selection criteria.

10.6.2 Local Capacity Ceiling and AnomalyClosureError

The second test probes the finite-observer cutoff independently of the linear heat slope. To make the notation explicit, define the Causal Point's local capacity ceiling by

$$C_{\text{local}} \equiv N_{\text{limit}} = \min \left(N_{\text{local}}, \frac{A_{\text{local}}}{4L_P^2 \ln 2} \right). \quad (256)$$

Metric or parameter	Null-model H_0 target	SHBT H_1 target	Selection threshold
Direct slope $\hat{\alpha}$	0 zJ/bit	6.6990 zJ/bit	Reject H_0 at $\geq 5\sigma$, $p < 10^{-6}$
Normalized slope β_1	0.000 ± 0.010	1.000 ± 0.010	Requires consistency with the selected model
Intercept β_0	6.699 zJ	0.000 ± 0.180 zJ after baseline subtraction	Diagnoses parasitic drive heating
Reduced chi-square χ_ν^2	$\chi_\nu^2 \gg 1$ for SHBT-like data	1.00 ± 0.05	Noise-consistent goodness of fit
Akaike information criterion	AIC_{H_0} baseline	$\Delta\text{AIC} =$ $\text{AIC}_{H_0} - \text{AIC}_{H_1} > 10$	Decisive preference for H_1

Table 26: OLS/MLE parameters and binary model-selection thresholds.

The experimental residual entropy budget is

$$R_{\text{entropy}}^{\text{exp}} = C_{\text{local}} - C_{\text{op}}. \quad (257)$$

For the singleton, background-calibrated GET sequence, $C_{\text{op}} = C_{\text{get}} = k$. Normal state resolution requires $R_{\text{entropy}}^{\text{exp}} \geq 0$; if $C_{\text{op}} > C_{\text{local}}$, the negative residual triggers the SHBT failure condition denoted **AnomalyClosureError**.

The controlled cutoff run imposes an artificial local ceiling

$$C_{\text{local}}^{\text{exp}} = N_{\text{limit}}^{\text{exp}} = 6 \text{ bits}, \quad |\mathcal{R}|_{\text{max}} = 2^6 = 64, \quad (258)$$

and then continues the physical decoder sweep through $k = 10$. SHBT predicts

$$\begin{cases} k \leq 6 : & \text{admissible readout,} \\ k > 6 : & \text{loss of readout fidelity, nonlinear thermal-variance spikes,} \\ & \text{and state-resolution failure (AnomalyClosureError).} \end{cases} \quad (259)$$

Standard quantum mechanics predicts continuous address decoding across the same artificial ceiling provided the hardware remains functional. A sharp, reproducible breakdown at $C_{\text{op}} > C_{\text{local}}^{\text{exp}}$ therefore tests the finite-register postulate, while its absence falsifies this cutoff prediction even if a nonzero calorimetric slope were observed.

11 Closure Synthesis and Conclusions

SHBT comprises five coupled sectors: modular completion of the boundary theory, entropy self-resolution into bulk metric data, topological anti-baryon state decoupling with passive stress-energy preservation, entropy-budgeted observer history, and address-scaled calorimetric falsification. These are not independent mechanisms; they are projections of the same finite closed boundary register. Once the completed partition function is modular invariant and the canonical branch arithmetic closes, the bulk tensor, the physical Hamiltonian constraint, the baryogenesis fixed point, and the Causal Point history rule all become consequences of the same closure data; the calorimetric experiment tests the observer-cost consequence directly.

The core closure chain is

$$\boxed{[M, S_\partial] = [M, T_\partial] = 0 \iff \Delta_{\text{fr}} = 0 \iff \mathcal{E}_{\mu\nu} = 0 \iff \Pi_{\text{phys}} H_{\text{total}} \Pi_{\text{phys}} = 0.} \quad (260)$$

Equivalently, $H_{\text{total}} = 0$ after restriction to the physical subspace. The first condition says that the torus partition function is a genuine modular invariant of the completed boundary CFT. The second says that the visible levels and parent completion level have no residual framing mismatch. The third says that the bulk closure tensor vanishes, so the SHBT topological Einstein equation is satisfied without adding an independent bulk correction. The fourth says that the static physical sector has no external time generator left over after projection. The canonical branch is therefore the branch on which boundary modular covariance, visible-sector anomaly closure, bulk geometry, and the Hamiltonian constraint are the same statement viewed through different interfaces.

The geometric part of the construction is summarized by the holographic RG map

$$\boxed{\nabla_{\mathcal{C}} S_\partial \longrightarrow \rho_E(i, j) \longrightarrow \ell_r(\tau) \longrightarrow g_{\mu\nu}(\tau).} \quad (261)$$

The boundary entropy gradient is first converted into the normalized entanglement density $\rho_E(i, j)$ on the finite modular-overlap coordinate set. The ordered resolved state is then compressed into the prime-lattice load vector $\ell_r(\tau)$, and the stabilized Gram projection of this load vector gives the effective metric slice $g_{\mu\nu}(\tau)$. In this sense geometry is not a continuous background into which the boundary theory is inserted. It is the rendered image of finite boundary information after entropy gradients have been normalized, ordered, compressed, and projected.

The matter-asymmetry sector closes by the topological anti-baryon state decoupling loop

$$\boxed{\rho_B^* = \frac{\mathcal{D}_{\bar{B}} \rho_B^* \mathcal{D}_{\bar{B}}^\dagger}{\text{Tr}(\mathcal{D}_{\bar{B}} \rho_B^* \mathcal{D}_{\bar{B}}^\dagger)}, \quad \mathcal{C}_{\bar{B}}(\rho_B^*) = 0, \quad \delta T_{\mu\nu}^{\text{passive}} = 0.} \quad (262)$$

At the fixed point, active anti-baryon color and electromagnetic render charge have been stripped from the visible channel, but the passive gravitational source has not been discarded. It is carried by the dark completion sector. The resulting baryon excess is therefore an asymmetry of rendered visible charge, not a violation of the completed stress-energy account. This is why the same branch that fixes $\Delta_{\text{fr}} = 0$ also enforces $\delta T_{\mu\nu}^{\text{passive}} = 0$: the Stinespring isometric state-projection map is allowed only because it is completion preserving.

The same modular ledger that fixes η_B also yields $\Omega_{\text{DM}}/\Omega_b \simeq 5.4$, identifying dark matter as the passive gravitational ghost of decoupled anti-baryons.

The observer sector supplies the final link in the same chain. A Causal Point does not read a pre-existing classical history from an infinite archive. It crystallizes a finite ordered sequence of admissible projections,

$$\boxed{\Pi_{\text{history}} = \Pi_{A_m, \iota_m} \cdots \Pi_{A_2, \iota_2} \Pi_{A_1, \iota_1}.} \quad (263)$$

Each factor is admitted only if the local entropy budget remains nonnegative. The apparent classical past is therefore the record that survives the ordered application of finite-register GET operations. The observer is part of the cosmic code rather than an external reader of it: the address A_n , outcome index ι_n , local hidden-state budget, and pointer packet are all internal boundary data.

The principal numerical retrodictions of the canonical branch can be collected as

$$\boxed{\Lambda_{\text{holo}} \simeq 1.09 \times 10^{-52} \text{ m}^{-2}, \quad N \simeq 3.31 \times 10^{122}, \quad \eta_B = \mathcal{O}(10^{-10}), \quad g_{\mu\nu} = g_{\mu\nu}(\nabla_C S_\partial).} \quad (264)$$

The first number is the holographic dark-energy curvature scale used by the topological Einstein equation. The second is the finite boundary bit budget associated with that scale. The third is the order of the visible baryon asymmetry produced by the state-decoupling loop, with the canonical value $\eta_B = 6.449923359416 \times 10^{-10}$. The fourth is the structural prediction: the effective metric is not fundamental input but a calculable output of the entropy-gradient pipeline.

Relative to standard holographic approaches, SHBT keeps the boundary-first logic but changes what is taken as primitive. It does not start from a large- N gauge theory with an asymptotically anti-de Sitter bulk and then ask for a semiclassical interior. It starts from a completed static boundary CFT, requires exact modular closure, and lets finite entropy data render the bulk metric. Relative to entropic-gravity pictures, SHBT does not infer gravity from entropy alone; it constrains the entropy flow by the modular invariant pairing matrix, the canonical branch arithmetic, and the finite register size. Relative to purely constraint-based quantum gravity, it does not stop at the formal statement that the Hamiltonian vanishes on physical states; it ties that statement to visible-sector levels, dark completion, baryogenesis, and observer memory.

The most direct tests of the framework are therefore consistency tests across sectors. The branch data must give integer locks $I_\ell^* = 6$ and $I_q^* = 13$, zero framing defect, a vanishing bulk closure tensor, and a finite physical-sector Hamiltonian constraint. The entropy pipeline must produce symmetric, trace-normalized, positive metric slices from the same boundary densities that define the loading sequence. The baryogenesis module must remove active visible anti-baryon charge while keeping passive stress-energy unchanged. The Causal Point module must crystallize only those histories whose retrieval cost fits inside the local entropy budget. A sub-10 mK implementation must then distinguish the predicted address slope $\hat{\alpha} = N_{\text{pulse}} k_B T \ln 2$ from the null slope and test the sharp failure condition at $C_{\text{op}} > C_{\text{local}}$. Any failure of one of these checks would invalidate the canonical SHBT closure chain rather than an auxiliary numerical routine alone.

Several limitations remain. The analysis establishes and verifies one canonical branch rather than a classification theorem for all modular completions. The dark completion sector is fixed by closure requirements but still needs a more explicit microscopic construction. The entropy-to-metric map is deterministic on the canonical branch, yet a full comparison with precision cosmological data would require a larger phenomenological layer: survey observables, perturbations, noise models, and parameter-estimation pipelines. The baryogenesis mechanism also needs extension from the fixed render-channel identity to a complete account of early-universe thermal history as seen by finite boundary observers. The calorimeter architecture and statistical thresholds specified here constitute a prospective experiment, not a reported detection; fabrication yield, calibrated excess-noise spectra, and blinded control data remain necessary before either hypothesis can be accepted. These requirements define the next tests needed for a quantitatively predictive research program.

The closure may be stated in code-theoretic terms. A finite boundary register, when forced to close under completed modular kernels, contains more than a list of states. It contains the rules for which visible charges can be rendered, which passive sources must be preserved, which entropy gradients become geometry, and which observer histories can crystallize. In this view spacetime is the readable projection of a finite holographic code; matter asymmetry is a render-channel fixed

point of that code; and observation is the entropy-budgeted act of selecting a consistent history from within the same code. SHBT is therefore static only at the level of the completed boundary. The observer-accessible dynamics is the internally crystallized image of that closed finite boundary information.

12 Code Availability

The reference implementation and audit scripts for the SHBT and precision cosmology calculations are maintained in the GitHub repository¹. Reproducibility is tied to the tagged repository release accompanying the published analysis. The publication-critical entry points and their physical roles are summarized in Table 27. This section provides the implementation bridge for the theory developed in Sections 1–11.

Software interface	Physical role and principal accessors
<code>shbt_simulator</code>	PyO3 module exposing the foundation simulator, report getters, and record classes
<code>StaticBoundary</code>	branch arithmetic, modular blocks, partition evaluation, entropy densities, dominant sequence, and self-resolution
<code>ShbtSimulator</code>	full audit orchestration through <code>ShbtSimulator.run_full_audit()</code>
<code>HolographicProjection / BulkMetricSlice</code>	load-vector construction, Euler flux, metric synthesis, spatial projection, and slice spectra
<code>CausalPoint</code>	light-cone construction, local property packets, finite-memory admissibility, and history crystallization
<code>AnomalyClosureError</code>	finite-capacity exception defined in <code>src/shbt/causal_point.rs</code> and exported through <code>src/lib.rs</code>
<code>LightConeSample, LocalPropertyPacket, CoordinateLogEntry, MemoryReport</code>	observer-record types supporting the Causal Point closure conditions
<code>BaryogenesisIdentity</code>	C_{sph} , $J_{\text{CP}}^{\text{topo}}$, projector rank, restoration scale, and η_B
<code>BaryogenesisOptimizer, FieldSimulation, BenchmarkDelta</code>	anti-baryon charge decoupling, passive-source preservation, and implementation profiling
<code>precision_cosmology.py</code>	completed ledger, Hubble loading, growth, cluster collapse, dark matter, BBN, ISW, neutrinos, chronometers, entropy, and GET costs
<code>examples/run_audit.py</code>	canonical foundation-audit entry point
<code>src/shbt/boundary.rs</code>	boundary branch, modular blocks, partition evaluator, and closure checks
<code>src/shbt/entropy_flow.rs</code>	entropy cascade and holographic metric projection
<code>src/shbt/baryogenesis.rs</code>	baryogenesis identity, κ_{D_5} , state decoupling, and benchmark comparison
<code>src/shbt/causal_point.rs</code>	Causal Point light cone, finite-memory checks, and history crystallization
<code>src/lib.rs</code>	PyO3 exports and <code>ShbtSimulator</code> orchestration

Table 27: Explicit physics-to-software bridge required to reproduce the paper-level closure tables.

The canonical foundation call is `report=shbt_simulator.ShbtSimulator().run_full_audit()`, followed by `audit=report.to_dict()`. The boundary branch and ledger accessors are `StaticBoundary.benchmark_branch`, `StaticBoundary.lepton_level`, `StaticBoundary.quark_level`, `StaticBoundary.parent_level`, `StaticBoundary.i_l_star`, `StaticBoundary.i_q_star`, `StaticBoundary.c_dark_residual`, and `StaticBoundary.c_dark`. The visible-block constants are `CHARGE_EMBEDDING` and `LOW_SU3_WEIGHTS`; central charges and modular data are exposed through `StaticBoundary.su2_central_charge(26)`, `StaticBoundary.su3_central_charge(8)`,

¹Immutable release tag: <https://github.com/syslown/shbt-precision/releases/tag/v1.0.0>; commit-sha: `main@head`.

StaticBoundary.build_su2_visible_block(), StaticBoundary.build_su3_visible_block(), StaticBoundary.s_boundary, StaticBoundary.t_boundary, and StaticBoundary.evaluate_z_boundary_py(tau_re, tau_im).

The boundary entropy map uses StaticBoundary.build_loading_density(), StaticBoundary.build_entanglement_density(), StaticBoundary.build_dominant_sequence(), StaticBoundary.entropy_self_resolution(), and StaticBoundary.derive_temporal_increment(). The metric map uses HolographicProjection.derive_load_vector(), HolographicProjection.metric_from_load_vector(), and HolographicProjection.project_static_block_to_bulk(). The exported slice fields are BulkMetricSlice.load_vector, BulkMetricSlice.euler_flux, BulkMetricSlice.metric_components, BulkMetricSlice.spatial_metric, and BulkMetricSlice.eigenvalues.

The foundation report getters are report.branch, report.framing_defect, report.modular_invariant, report.zero_energy_locked, report.projection_dimension_26_to_4, report.eta_b, report.stress_energy_preserved, and report.memory_all_passed. The boundary dictionary is audit['boundary_report'] with keys modular_S_commutator, modular_T_commutator, loading_normalized, entanglement_density_normalized, modular_invariant, integral_spin_closure, and all_passed. The projection dictionary audit['projection_report'] contains slice_count, projector_rank, symmetric, trace_normalized, and positive_definite; the slice list is audit['metric_slices'].

The baryogenesis record exposes BaryogenesisIdentity.sphaleron_coefficient, BaryogenesisIdentity.jarlskog_topological, BaryogenesisIdentity.Pi_rank, BaryogenesisIdentity.deltaPi_126_match, BaryogenesisIdentity.structural_exponent, BaryogenesisIdentity.modular_restoration_scale_gev, and BaryogenesisIdentity.heavy_neutrino_to_planck_ratio, and BaryogenesisIdentity.eta_b. The corresponding dictionary is audit['baryogenesis_identity']. Active charge decoupling and profiling use BaryogenesisOptimizer.cpu_cycle_weight(), BaryogenesisOptimizer.derender_antibaryon_charges(), and BaryogenesisOptimizer.run_benchmark(particle_count); the comparison is audit['benchmark_delta']. Its BenchmarkDelta keys are standard, optimized, cpu_cycle_delta, operation_delta, memory_delta_bytes, elapsed_delta_s, cpu_cycle_reduction_fraction, operation_reduction_fraction, memory_reduction_fraction, and stress_energy_preserved. The nested FieldSimulation records expose field_name, particle_count, cpu_cycle_weight, operation_count, memory_bytes, elapsed_s, peak_traced_bytes, and checksum.

The observer interface uses CausalPoint.build_past_light_cone(), CausalPoint.compute_property_packets(), and CausalPoint.crystallize_history(). Its dictionary audit['memory_report'] exposes local_available_bits, hidden_bits, entropy_limit_bits, past_light_cone_samples, and property_packets; crystallized records are returned in audit['history_entries']. GET requests retain the exact fields observable_name, amplitude, c_vis, and c_dark.

The precision-cosmology entry points are load_completed_ledger, load_default_constants, loading_fraction_ode, compute_loading_fraction, compute_entropy_debt, entropy_debt_uplift_factor, h0_local, loading_amplitude, lock_rate, h0_redshift_dependent, shbt_hubble_rate, growth_ode_system, compute_growth_suppression, compute_cluster_collapse,

`compute_dark_matter_density`, `compute_dm_baryon_ratio`, `dark_matter_equation_of_state`, `isw_residual`, `bbn_stability_check`, `neutrino_hierarchy_masses`, `get_measurement_cost`, `collapse_index`, and `build_precision_cosmology_report`. The private helper functions required for exact reproduction are `compute_kappa_d5`, `_cpl_template`, and `_bbn_components`. Default-constant fields used by the calculations are `lambda_holo_si_m2`, `n_sat`, and `h0_cmb`.

The consolidated dictionary keys are `completed_ledger`, `h0_local_km_s_mpc`, `redshift_ladder`, `lightcone_entropy_debt`, `growth_suppression`, `cpl_template`, `cluster_collapse`, `dark_matter`, `isw_stability`, `bbn_stability`, `neutrino_hierarchy`, `cosmic_chronometer_validation`, `thermodynamic_arrow`, and `summary_table_17`. The dark-matter subrecord preserves `rho_DM_kg_m3`, `Omega_DM`, `Omega_b`, `Omega_DM_over_Omega_b`, `w_DM`, and `abundance_ratio`.

Implementation profiling is intentionally separated from the theoretical tables. The normalized output of `BaryogenesisOptimizer.run_benchmark(particle_count)` stored in `audit['benchmark_delta']` is

Profile quantity	Standard render	Projected SHBT render	Reduction fraction
CPU-cycle weight	1.0	0.5	0.5
Operation count	1.0	0.0015333147477	0.9984666852523
Memory bytes	1.0	0.0058214747736093	0.9941785252263907

These are software execution diagnostics only; the passive stress-energy condition remains `stress_energy_preserved=True` in both descriptions.

The repository-root reproduction commands are:

```
python examples/run_audit.py
python precision_cosmology.py --run-tests
python precision_cosmology.py --json
python shbt_simulate.py --mode cosmology
python shbt_simulate.py --mode cosmology --format csv --output cosmo
cargo test
cargo build --release
```

The optional Python binding checks use `pytest tests/test_simulator.py -q`, and a release wheel can be built with `maturin build -release`.

Appendices: Modular Framing and Quantum-Dimension Derivations

In the appendices, c_{dark} denotes the completed ledger $c_{\text{dark}}^{\text{comp}}$; the residual ledger is written explicitly as $c_{\text{dark}}^{\text{res}}$.

A Modular Framing Phase Proof

In Static Holographic Boundary Theory (SHBT), the fundamental boundary object is the completed torus partition function $Z_{\partial}(\tau, \bar{\tau})$, defined over the complex upper half-plane $\tau = \tau_1 + i\tau_2 \in \mathbb{H}$ by

$$Z_{\partial}(\tau, \bar{\tau}) = \sum_{I,J} M_{IJ} \chi_I(\tau) \bar{\chi}_J(\bar{\tau}) = \chi(\tau)^{\dagger} M \chi(\tau), \quad (265)$$

where $\chi(\tau) = (\chi_0(\tau), \chi_1(\tau), \dots)^T$ represents the formal character vector of the completed boundary conformal field theory (CFT), and $M_{IJ} \in \mathbb{Z}_{\geq 0}$ is the non-negative modular framing pairing matrix normalized such that $M_{00} = 1$.

A.1 Modular Group Transformation Kernels

The modular group $PSL(2, \mathbb{Z}) \cong SL(2, \mathbb{Z})/\{\pm \mathbb{I}\}$ acts on the modular parameter $\tau \in \mathbb{H}$ via fractional linear transformations generated by:

$$S : \tau \mapsto -\frac{1}{\tau}, \quad T : \tau \mapsto \tau + 1, \quad (266)$$

subject to the canonical algebraic relations $S^2 = (ST)^3 = C$, where C is the charge conjugation operator satisfying $C^2 = \mathbb{I}$. The transformation of the completed character vector under these generators is mediated by the unitary boundary modular representation matrices S_{∂} and T_{∂} :

$$\chi_I \left(-\frac{1}{\tau} \right) = \sum_J (S_{\partial})_{IJ} \chi_J(\tau), \quad \chi_I(\tau + 1) = \sum_J (T_{\partial})_{IJ} \chi_J(\tau). \quad (267)$$

The completed boundary CFT Hilbert space factors tensorially into visible and completion sectors:

$$\mathcal{H}_{\partial}^{\text{comp}} = \mathcal{H}_{SU(2)_{k_l}} \otimes \mathcal{H}_{SU(3)_{k_q}} \otimes \mathcal{H}_{SO(10)_K} \otimes \mathcal{H}_{\text{dark}}, \quad (268)$$

inducing a Kronecker factorized structure on the boundary modular transformation kernels:

$$S_{\partial} = S_{SU(2)_{k_l}} \otimes S_{SU(3)_{k_q}} \otimes S_{SO(10)_K} \otimes S_{\text{dark}}, \quad (269)$$

$$T_{\partial} = T_{SU(2)_{k_l}} \otimes T_{SU(3)_{k_q}} \otimes T_{SO(10)_K} \otimes T_{\text{dark}}. \quad (270)$$

For an arbitrary affine Wess-Zumino-Witten (WZW) factor G_k , highest weight modules are parameterized by integrable highest weights $\lambda \in P_+^k$. The central charge c_{G_k} and conformal weight h_{λ} are given by:

$$c_{G_k} = \frac{k \dim G}{k + g^{\vee}}, \quad h_{\lambda} = \frac{(\lambda, \lambda + 2\rho)}{2(k + g^{\vee})}, \quad (271)$$

where $g^{\vee}$ is the dual Coxeter number and ρ is the Weyl vector. The explicit constituent modular kernels are defined as follows:

1. $SU(2)_{k_l}$ Sector: Integrable labels are integers $a \in \{0, 1, \dots, k_l\}$. The central charge is $c_{SU(2)_{k_l}} = \frac{3k_l}{k_l+2}$, and conformal weights are $h_a = \frac{a(a+2)}{4(k_l+2)}$. The modular kernels read:

$$(S_{SU(2)_{k_l}})_{a,b} = \sqrt{\frac{2}{k_l+2}} \sin\left(\frac{\pi(a+1)(b+1)}{k_l+2}\right), \quad (272)$$

$$(T_{SU(2)_{k_l}})_{a,b} = \delta_{a,b} \exp\left[2\pi i \left(\frac{a(a+2)}{4(k_l+2)} - \frac{c_{SU(2)_{k_l}}}{24}\right)\right]. \quad (273)$$

2. $SU(3)_{k_q}$ Sector: Integrable labels are Dynkin pairs (p, q) with $p, q \geq 0$ and $p + q \leq k_q$. The central charge is $c_{SU(3)_{k_q}} = \frac{8k_q}{k_q+3}$, and conformal weights are $h_{(p,q)} = \frac{p^2+q^2+pq+3p+3q}{3(k_q+3)}$. Let $v(p, q) = \left(\frac{2p+q}{3}, \frac{-p+q}{3}, \frac{-p-2q}{3}\right)$ denote the weight vector in the trace-free coordinate system, with $\rho = (1, 0, -1)$. The modular kernels are:

$$(S_{SU(3)_{k_q}})_{(p,q),(r,s)} = \frac{i^3}{\sqrt{3}(k_q+3)} \sum_{w \in S_3} \text{sgn}(w) \exp\left[-\frac{2\pi i}{k_q+3} \left(w(v(p, q) + \rho), v(r, s) + \rho\right)\right], \quad (274)$$

$$(T_{SU(3)_{k_q}})_{(p,q),(r,s)} = \delta_{(p,q),(r,s)} \exp\left[2\pi i \left(\frac{p^2+q^2+pq+3p+3q}{3(k_q+3)} - \frac{c_{SU(3)_{k_q}}}{24}\right)\right]. \quad (275)$$

3. $SO(10)_K$ Parent Completion Sector: The Lie algebra is D_5 with $g^\vee = 8$, rank 5, and dimension 45. The central charge is $c_{SO(10)_K} = \frac{45K}{K+8}$. For integrable highest weights $\Lambda, M \in P_+^K(D_5)$, the modular kernels are:

$$(S_{SO(10)_K})_{\Lambda, M} = \frac{i^{20}}{2(K+8)^{5/2}} \sum_{w \in W(D_5)} \text{sgn}(w) \exp\left[-\frac{2\pi i}{K+8} \left(w(\Lambda + \rho), M + \rho\right)\right], \quad (276)$$

$$(T_{SO(10)_K})_{\Lambda, M} = \delta_{\Lambda, M} \exp\left[2\pi i \left(\frac{(\Lambda, \Lambda + 2\rho)}{2(K+8)} - \frac{45K}{24(K+8)}\right)\right]. \quad (277)$$

A.2 Proof of Matrix Commutation Relations

Exact modular invariance of the physical torus partition function $Z_\partial(\tau, \bar{\tau})$ requires $Z_\partial(\tau+1, \bar{\tau}+1) = Z_\partial(\tau, \bar{\tau})$ and $Z_\partial(-1/\tau, -1/\bar{\tau}) = Z_\partial(\tau, \bar{\tau})$ for all $\tau \in \mathbb{H}$. In matrix notation:

$$Z_\partial(\tau+1, \bar{\tau}+1) = \chi(\tau)^\dagger T_\partial^\dagger M T_\partial \chi(\tau) = \chi(\tau)^\dagger M \chi(\tau), \quad (278)$$

$$Z_\partial\left(-\frac{1}{\tau}, -\frac{1}{\bar{\tau}}\right) = \chi(\tau)^\dagger S_\partial^\dagger M S_\partial \chi(\tau) = \chi(\tau)^\dagger M \chi(\tau). \quad (279)$$

Because S_∂ and T_∂ are unitary transformations ($S_\partial^\dagger = S_\partial^{-1}$ and $T_\partial^\dagger = T_\partial^{-1}$), equations (278) and (279) hold generically if and only if:

$$[M, S_\partial] = 0 \quad \text{and} \quad [M, T_\partial] = 0. \quad (280)$$

Proof. Evaluating elementwise: 1. **Evaluation of $[M, T_\partial] = 0$:** Since $(T_\partial)_{IJ} = \theta_I \delta_{IJ}$ with $\theta_I \equiv \exp[2\pi i(h_I - \frac{c_I}{24})]$, one obtains $([M, T_\partial])_{IJ} = M_{IJ}(\theta_J - \theta_I) = 0$. If $M_{IJ} \neq 0$, then $\theta_I = \theta_J$, yielding the integral-spin condition:

$$h_I - h_J - \frac{c_I - c_J}{24} \in \mathbb{Z}. \quad (281)$$

2. **Evaluation of $[M, S_\partial] = 0$:** In a center-resolved block where $M_{IJ} = m_I \delta_{IJ}$ is diagonal, $(m_I - m_J)(S_\partial)_{IJ} = 0$. Since $(S_\partial)_{IJ} \neq 0$ across integrable representations, $m_I = m_J$. Normalizing $M_{00} = 1$ fixes $M = \mathbb{I}$, satisfying $[\mathbb{I}, S_\partial] = 0$ unconditionally. $\square$

A.3 Proof of Vanishing Canonical Framing Defect Δ_{fr}

The center-lift embedding locks connect parent level K to visible levels k_l and k_q via center quotient lifts:

$$I_l^* \equiv \frac{K}{2k_l}, \quad I_q^* \equiv \frac{K}{3k_q}. \quad (282)$$

The scalar framing defect $\Delta_{\text{fr}}(K, k_l, k_q)$ is defined by:

$$\Delta_{\text{fr}}(K, k_l, k_q) \equiv \max \left(\left\| \frac{K}{2k_l} \right\|_{\mathbb{Z}}, \left\| \frac{K}{3k_q} \right\|_{\mathbb{Z}} \right), \quad (283)$$

where $\|x\|_{\mathbb{Z}} \equiv \min_{n \in \mathbb{Z}} |x - n|$.

Theorem 1. *On the canonical branch $\mathfrak{b}_* = (k_l, k_q, K) = (26, 8, 312)$, the framing defect evaluates strictly to zero:*

$$\Delta_{\text{fr}}(312, 26, 8) = 0. \quad (284)$$

Proof. Evaluating center lifts I_l^* and I_q^* on $(26, 8, 312)$:

$$I_l^* = \frac{312}{2 \times 26} = \frac{312}{52} = 6 \in \mathbb{Z}, \quad (285)$$

$$I_q^* = \frac{312}{3 \times 8} = \frac{312}{24} = 13 \in \mathbb{Z}. \quad (286)$$

Hence, $\|6\|_{\mathbb{Z}} = 0$ and $\|13\|_{\mathbb{Z}} = 0$, giving $\Delta_{\text{fr}}(312, 26, 8) = \max(0, 0) = 0$. This forces the bulk closure tensor $\mathcal{E}_{\mu\nu} = 0$ and satisfies $\Pi_{\text{phys}} H_{\text{total}} \Pi_{\text{phys}} = 0$. $\square$

B Boundary Quantum Dimension & Character Derivations

B.1 Character Ratio Asymptotics and the Verlinde Limit

Definition 1. *The quantum dimension d_i of a primary field i is defined by the asymptotic character ratio limit as $q \rightarrow 1^-$ along $\tau = iy$ ($y \rightarrow 0^+$):*

$$d_i \equiv \lim_{q \rightarrow 1^-} \frac{\chi_i(q)}{\chi_0(q)} = \lim_{y \rightarrow 0^+} \frac{\chi_i(iy)}{\chi_0(iy)}. \quad (287)$$

Theorem 2. For any primary field i in a modular invariant rational CFT:

$$d_i = \frac{S_{i0}}{S_{00}}. \quad (288)$$

Proof. Under $\tau \mapsto -1/\tau$, $y \rightarrow 0^+$ maps to dual modular parameter $\tilde{q} = e^{-2\pi/y} \rightarrow 0$. Transforming characters via $\chi_i(iy) = \sum_j S_{ij} \chi_j(i/y)$ and expanding $\chi_j(i/y) = \tilde{q}^{-c/24}(\delta_{j0} + \mathcal{O}(\tilde{q}))$:

$$d_i = \lim_{\tilde{q} \rightarrow 0} \frac{\tilde{q}^{-c/24} \left(S_{i0} + \sum_{j \neq 0} S_{ij} \mathcal{O}(\tilde{q}^{h_j}) \right)}{\tilde{q}^{-c/24} \left(S_{00} + \sum_{j \neq 0} S_{0j} \mathcal{O}(\tilde{q}^{h_j}) \right)} = \frac{S_{i0}}{S_{00}}. \quad (289)$$

□

B.2 Explicit Evaluation across Visible Affine Sectors

On $(k_l, k_q) = (26, 8)$:

1. $SU(2)_{26}$ Sector ($k_l + 2 = 28$):

$$d_a^{SU(2)} = \frac{\sin\left(\frac{(a+1)\pi}{28}\right)}{\sin\left(\frac{\pi}{28}\right)}. \quad (290)$$

Evaluations: $d_0 = 1$, $d_1 = 2 \cos(\pi/28) \approx 1.98742442$, $d_{22} = \frac{\sin(5\pi/28)}{\sin(\pi/28)} \approx 4.75179356$, $d_{23} = \frac{\sin(4\pi/28)}{\sin(\pi/28)} \approx 3.87519108$, $d_{26} = 1$.

2. $SU(3)_8$ Sector ($k_q + 3 = 11$):

$$d_{(p,q)}^{SU(3)} = \frac{\sin\left(\frac{(p+1)\pi}{11}\right) \sin\left(\frac{(q+1)\pi}{11}\right) \sin\left(\frac{(p+q+2)\pi}{11}\right)}{\sin^2\left(\frac{\pi}{11}\right) \sin\left(\frac{2\pi}{11}\right)}. \quad (291)$$

Evaluations: $d_{(0,0)} = 1$, $d_{(1,0)} = d_{(0,1)} = \frac{\sin(3\pi/11)}{\sin(\pi/11)} \approx 2.68250707$, $d_{(1,1)} = \frac{\sin(2\pi/11) \sin(4\pi/11)}{\sin^2(\pi/11)} \approx 6.19584416$.

B.3 Retained D_5 Boundary-Cell Measure

The retained D_5 measure is the trace-normalized overlap of the completed spinor cell with the visible boundary block. The canonical branch fixes this dimensionless overlap to

$$\boxed{\kappa_{D_5} = 0.988769793998.} \quad (292)$$

This is the retained quantum-dimension weight entering the topological Jarlskog residue and the finite-register neutrino scale.

The corresponding normal-hierarchy eigenvalues are

$$m_{\nu,1} = \kappa_{D_5} M_P N_{\text{sat}}^{-1/4} = 2.829630635353 \times 10^{-3} \text{ eV}, \quad (293)$$

$$m_{\nu,2} = \sqrt{m_{\nu,1}^2 + \Delta m_{21}^2} \simeq 8.92 \times 10^{-3} \text{ eV},$$

$$m_{\nu,3} = \sqrt{m_{\nu,1}^2 + \Delta m_{31}^2} \simeq 5.01 \times 10^{-2} \text{ eV}. \quad (294)$$

B.4 Exact Dark Central Charge Ledger Proof

Theorem 3. *Visible central charge:* $c_{\text{vis}} = c_{SU(2)_{26}} + c_{SU(3)_8} = \frac{39}{14} + \frac{64}{11} = \frac{1325}{154} \approx 8.603896103896$.

Theorem 4. *The residual dark ledger $c_{\text{dark}}^{\text{res}} = \frac{834433}{362670}$ and completed dark ledger $c_{\text{dark}}^{\text{comp}} = \frac{1197103}{362670}$ satisfy $c_{\text{dark}}^{\text{comp}} - c_{\text{dark}}^{\text{res}} = 1$. Both share the denominator $D = 362670 = 2 \times 3 \times 5 \times 7 \times 11 \times 157$ with c_{vis} , reducing via $\text{gcd} = 22 = 2 \times 11$ to $c_{\text{tot}}^{\text{res}} = \frac{179764}{16485}$ and $c_{\text{tot}}^{\text{comp}} = \frac{196249}{16485}$.*

$$c_{\text{dark}}^{\text{comp}} - c_{\text{dark}}^{\text{res}} = \frac{1197103 - 834433}{362670} = 1. \quad (295)$$

Proof. Since $154 = 2 \times 7 \times 11$ divides 362670 with multiplier $K_{\text{mult}} = 3 \times 5 \times 157 = 2355$:

$$c_{\text{tot}}^{\text{res}} = \frac{1325 \times 2355 + 834433}{362670} = \frac{3954808}{362670} = \frac{3954808/22}{362670/22} = \frac{179764}{16485}, \quad (296)$$

$$c_{\text{tot}}^{\text{comp}} = \frac{1325 \times 2355 + 1197103}{362670} = \frac{4317478}{362670} = \frac{4317478/22}{362670/22} = \frac{196249}{16485}. \quad (297)$$

The reduced denominator $16485 = 3 \times 5 \times 7 \times 157$ reflects prime conductor 157. $\square$

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